

# Cascade criticality, therapeutic window emergence, and pre-symptomatic early-warning signals in a connectome-based neurodegeneration framework: an ALS-motivated study on the *C. elegans* motor circuit

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**CRITICAL DISCLAIMERS:** This is a computational modelling study using the *C. elegans* 61-neuron motor connectome as a substrate for studying abstract degeneration dynamics. It is **not a clinical model of ALS**, does not use human neural circuitry, and cannot make predictions about human patients. The *C. elegans* motor system differs fundamentally from the human corticospinal-motor neuron system affected in ALS. All model parameters are chosen for dynamical plausibility, not biological calibration to measured quantities in any organism. Threshold values, timescales, and parameter ranges are illustrative. Clinical time-mapping (e.g., “symptom onset at step 200”) is an analogy for communication purposes only. **All results are hypothesis-generating and require experimental validation before any translational significance can be attributed to them.**

## Abstract

Amyotrophic lateral sclerosis (ALS) is characterised by progressive motor neuron degeneration whose mechanisms remain incompletely understood. A key open question is whether degeneration is a continuous process or involves discrete tipping points, and whether the timing of therapeutic intervention is intrinsically constrained by circuit-level dynamics. Here we present a multi-phase computational study using the empirical *C. elegans* 61-neuron motor connectome as a tractable substrate for exploring how biophysical feedback loops, network topology, and aggregation dynamics interact to produce catastrophic collapse.

We implement a nonlinear biophysical cascade (aggregation → mitochondrial damage → ATP collapse → glutamate excitotoxicity → calcium overload → oxidative stress → aggregation) on the full directed connectome (61 neurons, 127 synapses) and characterise 500 parameter configurations across three dynamical regimes: stable (14.4%), critical (76.4%), and runaway (9.2%). A three-part falsification criterion reduces the initial 44% null-model false-positive rate to 0%, identifying 247 configurations with genuine triphasic degeneration: a pre-symptomatic silent phase,

rapid collapse, and arrested plateau. Phase 14 stress-testing reveals a hierarchy of dynamical robustness: spatial coherence and strict tipping-point structure collapse under severe perturbation ( $>30\%$  edge dropout, vulnerability  $\sigma > 0.5$ ), while the relative ordering of degeneration aggressiveness remains fully preserved (Spearman  $r = 1.000$ ) even under conditions that destroy all other dynamical structure.

Aggregation suppression therapy consistently outperforms four other therapeutic strategies. The therapeutic window follows a sharp linear boundary in (strength, start-time) space:  $\text{max\_start\_t} = 425 \times \text{strength} - 237$  for the index configuration, generalising to mean  $252 \times \text{strength} - 107$  across 17 of 20 tested configurations ( $R^2 = 0.83$ ). Robustness testing reveals two disease subtypes differing in aggression and window shape. An ablation study demonstrates that topological load redistribution constitutes an independent, aggregation-independent escape mechanism capable of complete collapse in isolation. All principal findings were robust to four categories of biological noise (synaptic weight variation, vulnerability score perturbation, connectome edge dropout, and per-neuron aggregation heterogeneity), with all metrics changing by less than 4% relative to unperturbed baselines.

Round 2 extended the framework to topology resilience and proof-of-concept early-warning signal analysis. Sparse-chain topology improved cascade resilience ( $\text{RES} = 0.815$  vs. baseline  $\text{RES} = 0.800$ ), while triangle-rich feedback loops exhibit a critical safety threshold at  $\sim 60\%$  density beyond which cascade vulnerability increases; no topology modification widened the therapeutic window, confirming that circuit architecture and drug timing are independent intervention axes. Three pre-symptomatic biomarkers measured at  $t = 50$  (before any neuron death) achieved 87% simulated subtype classification accuracy (Phase R2.8 TWE pipeline) and achieved 86.8% of the oracle-defined benefit within this simulation framework, with within-subtype parameter heterogeneity — not stochastic noise — identified as the dominant irreducible error source.

These results generate testable hypotheses about the necessity of pre-symptomatic intervention, the primacy of protein aggregation seeding as a disease driver, and the proof-of-concept potential of simulated model-derived early-warning proxies for hypothesis-generating therapeutic triage, while underscoring that all findings remain model-specific and require experimental corroboration.

*Keywords:* neurodegeneration model; connectome; tipping points; therapeutic window; protein aggregation; ALS; *C. elegans*; model-derived early-warning proxies; topology resilience; cascade criticality

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# 1 Introduction

## 1.1 Background and motivation

A central challenge in computational neurodegeneration modelling is distinguishing genuine dynamical phenomena — tipping points, therapeutic windows, disease subtypes — from model artifacts. Here, using ALS as a motivating biological context and the *C. elegans* motor connectome as a principled network substrate, we develop and stress-test a multi-phase framework for studying cascade criticality and intervention timing in biophysically-coupled network degeneration models.

Amyotrophic lateral sclerosis is a fatal neurodegenerative disease characterised by progressive loss of both upper and lower motor neurons, leading to paralysis and death, typically within two to five years of symptom onset [van Es et al., 2017]. Despite advances in understanding molecular mechanisms — including TDP-43 pathology [Neumann et al., 2006], C9ORF72 repeat expansions [DeJesus-Hernandez et al., 2011], and SOD1 misfolding [Rosen et al., 1993] — the disease remains almost universally fatal, with only modest survival benefits from current therapeutics (riluzole, edaravone, sodium phenylbutyrate/taurursodiol).

A recurring clinical observation is that significant neurodegeneration may precede symptom onset by months to years [Turner et al., 2012, Benatar et al., 2019]. The ATLAS trial of tofersen, an antisense oligonucleotide targeting SOD1, demonstrated that pre-symptomatic intervention in biomarker-positive carriers could delay phenoconversion [Benatar et al., 2023]. This raises a fundamental theoretical question: is there an inherent, mechanistically-determined therapeutic window, and if so, what determines its width?

Separately, it is increasingly appreciated that ALS is heterogeneous [Al-Chalabi et al., 2017]: patients vary markedly in onset site, progression rate, extramotor involvement, and survival. Whether this heterogeneity reflects distinct disease subtypes with qualitatively different dynamics, or a continuum of the same underlying process, has implications for patient stratification and clinical trial design.

## 1.2 The *C. elegans* connectome as a model substrate

*Caenorhabditis elegans* has a fully mapped nervous system [White et al., 1986, Cook et al., 2019] comprising 302 neurons with complete synaptic connectivity. Its motor circuit — approximately 61 neurons with 127 directed synapses — is compact enough for exhaustive computational analysis while retaining genuine biological wiring rather than a synthetic graph approximation. The motor neurons include dorsal and ventral A-, B-, and D-type motor neurons (DA, DB, DD, VA, VD) with well-characterised vulnerability profiles to cellular stress.

We use this connectome not as a model of *C. elegans* neurobiology per se, but as a principled network substrate on which to impose ALS-relevant biophysical dynamics. This approach allows us to separate the contributions of network topology from biochemical feedback strength, in a system where topology is empirically determined rather than assumed.

**We emphasise that this is not a model of ALS in nematodes or humans.** *C. elegans* does not develop ALS. Its motor system differs from the human corticospinal-motor neuron system in every relevant respect: scale, architecture, cell type, and disease susceptibility. All results should be interpreted as dynamical properties of an abstract model on a biologically-derived graph.

### 1.3 Gaps addressed and contributions

This study addresses three gaps in computational ALS/neurodegeneration modelling:

1. **Tipping point falsification:** Existing network neurodegeneration models rarely subject claimed tipping points to rigorous null-model control. We develop and apply a three-part falsification criterion that reduces a 44% null-model false-positive rate to 0%.
2. **Topology vs. dynamics:** Whether degeneration patterns in connectome models are primarily driven by the specific wiring or by the underlying biochemical dynamics is rarely directly tested. We do so across five network types (Phases 5–14) and nine structural variants (Round 2).
3. **Therapeutic window quantification and proof-of-concept early-warning signal analysis:** We map the exact (strength, timing) boundary separating prevention from failure, characterise two disease subtypes, and — in Round 2 — demonstrate that pre-symptomatic biomarkers at the  $t = 50$  equivalent can assign intervention strategy with 87% subtype accuracy and 86.8% oracle efficiency.

## 2 Methods

### 2.1 Connectome data

The motor connectome was derived from the *C. elegans* physical wiring diagram [White et al., 1986, Cook et al., 2019]. We use 61 neurons with 127 directed chemical synapses in the motor circuit. Synaptic weights are derived from synapse counts. Vulnerability scores — representing each neuron’s intrinsic susceptibility to aggregation-driven damage — are assigned based on neuron type: motor neurons (DA, DB, DD, VA, VD) receive scores of 0.65–1.00 reflecting their established sensitivity to cellular stress.

**Parameter range justification.** Parameter ranges were selected to span subcritical, critical, and supercritical dynamical regimes rather than to match biological measurements. `aggregationAmplification` spans  $[0.05, 20.0]$  (log-uniform) to cover the full range from near-zero cascade to runaway collapse; the other six parameters were set to ranges wide enough that the critical regime occupies a substantial fraction of parameter space (76.4% of 500 sampled configurations), ensuring sufficient data for subtype and boundary analysis. These ranges are not intended as estimates of biological quantities in any organism.

The connectome does not include the DVA interneuron, which appears in simplified circuit diagrams but is absent from the canonical motor circuit reconstruction; the AVDL interneuron (index 12) serves as a structural proxy for high-connectivity interneuron function in relevant analyses. All simulation phases are interactively visualisable at <https://als-connectome-dashboard.vercel.app>.

### 2.2 Biological state variables

Each neuron  $i$  carries four continuous state variables updated at each timestep  $\Delta t = 1$ :

- **Health**  $h_i \in [0, 1]$ : fractional functional integrity; neuron is dead when  $h_i < 0.15$ .

- **Aggregation load**  $a_i \in [0, \infty)$ : normalised misfolded protein burden, analogous to TDP-43 or SOD1 aggregate density.
- **ATP**  $e_i \in [0, 1]$ : relative bioenergetic status; irreversible collapse locks at  $e_i = 0$  once both  $e_i < \theta_{\text{ATP}}$  and  $a_i > \theta_{\text{irrev}}$  simultaneously.
- **Glutamate drive**  $g_i$ : synaptic glutamate accumulation arising from reuptake failure under ATP depletion.

These variables are updated each step through coupled differential equations encoding the excitotoxic cascade described below.

## 2.3 Excitotoxic feedback cascade

The model implements six coupled feedback mechanisms forming a closed positive-feedback loop:

```

aggregation
→ mitochondrial damage    (mitochondrialFragility)
→ ATP collapse             (atpCollapseThreshold)
→ glutamate accumulation  (glutamateSensitivity × reuptake failure)
→ Ca2+ overload          (calciumStressGain × NMDA activation)
→ oxidative stress         (ROS from Ca2+)
→ aggregation seeding     (oxidativeFeedback) ← closes the loop

```

The four state variables evolve according to:

$$\frac{da_i}{dt} = \alpha \cdot \text{aggAmp} \cdot a_i(1 - a_i) + \beta \sum_{j \in \mathcal{N}(i)} a_j \cdot w_{ji} + \gamma \cdot \text{ROS}_i \quad (1)$$

$$\frac{de_i}{dt} = \rho_e(1 - e_i) - \kappa \cdot a_i - \delta \cdot \text{load}_i \quad (2)$$

$$\frac{dg_i}{dt} = \sigma \cdot (1 - e_i) - \lambda \cdot \text{astro}_i \cdot g_i \quad (3)$$

$$\frac{dh_i}{dt} = -\eta \cdot \text{stress}_i + \mu \cdot e_i \cdot (1 - a_i) \quad (4)$$

where  $\text{stress}_i = w_1 \cdot \text{overload}_i + w_2 \cdot a_i + w_3 \cdot g_i$ ,  $\mathcal{N}(i)$  is the set of neurons projecting to neuron  $i$ , and  $w_{ji}$  is the synaptic weight from  $j$  to  $i$ . Parameters  $\alpha, \beta, \gamma, \rho_e, \kappa, \delta, \sigma, \lambda, \eta, \mu$  correspond to the named simulation parameters; full values in Supplementary Table S1.

**Irreversibility switch:** when a neuron simultaneously satisfies  $e_i < \theta_{\text{ATP}}$  and  $a_i > \theta_{\text{irrev}}$ , its **irreversible** flag is set. From this point, health can only decrease; the neuron is committed to degeneration. This models the clinical observation that neurons past a bioenergetic threshold cannot recover energy homeostasis.

**Aggregation seeding and spread:** The aggregation term drives both intrinsic growth (within a neuron) and prion-like spread (to synaptic neighbours). The scaling factor **aggregationAmplification** acts as a bifurcation parameter: below a critical value, the loop gain is insufficient to sustain spreading; above it, the cascade becomes self-amplifying. Log-uniform sampling is applied to this parameter and to **glutamateSensitivity** and **oxidativeFeedback**, as these span multiple orders of magnitude in biological systems.

Seven free parameters are explored (Table 1). (*Parameter names use camelCase as implemented in the simulation code.*)

Table 1: Model parameters, ranges, and biological analogues.

| Parameter                | Range         | Sampling    | Biological analogue                      |
|--------------------------|---------------|-------------|------------------------------------------|
| aggregationAmplification | [0.05, 20.0]  | log-uniform | TDP-43/FUS prion-like seeding efficiency |
| mitochondrialFragility   | [0.3, 8.0]    | uniform     | PINK1/Parkin pathway, Complex I deficit  |
| atpCollapseThreshold     | [0.05, 0.7]   | uniform     | Bioenergetic point of no return          |
| glutamateSensitivity     | [0.0005, 0.1] | log-uniform | EAAT2 glutamate transporter expression   |
| calciumStressGain        | [0.02, 5.0]   | uniform     | NMDA receptor density                    |
| oxidativeFeedback        | [0.0005, 0.5] | log-uniform | SOD1 activity / antioxidant reserve      |
| recoveryIrreversibility  | [0.2, 0.99]   | uniform     | Autophagic clearance capacity            |

## 2.4 Regime classification

500 parameter configurations were run for 500 timesteps each. Three regimes are defined:

- **Stable:**  $>50$  neurons alive at  $t = 500$  (network resists degeneration).
- **Critical:**  $\geq 10$  alive at  $t = 200$  AND  $\leq 50$  alive at  $t = 500$  (sustained, partial collapse).
- **Runaway:**  $<10$  alive by  $t = 200$  (rapid near-complete collapse).

The critical regime is the focus of subsequent analysis; it corresponds to sustained intermediate degeneration most consistent with clinical ALS progression.

All results are reported in simulation steps only. Day equivalents in the Discussion are order-of-magnitude analogies based on a single-anchor calibration (step 200  $\approx$  ALS symptom onset  $\approx$  15 months) and must not be interpreted as calibrated predictions.

## 2.5 Tipping point criterion — falsification approach

An initial single-threshold criterion (alive  $< 55$  at any step) produced a 44% false-positive rate against an independent null model in which neurons decline via uncoupled random walks with no synaptic coupling or biochemical feedback. This prompted development of a three-part strict criterion:

- **C1 — Slope:** peak 10-step neuron death rate  $> 4$  (sudden acceleration).
- **C2 — Spatial coherence:** Pearson  $r$  between each neuron’s vulnerability score and its death step  $> 0.30$  (death order correlated with vulnerability, as expected from synaptic spread).
- **C3 — Silent phase:** first neuron death after step 50 (clinically meaningful pre-symptomatic window).



All three conditions must be satisfied simultaneously. Under this criterion, the null model false-positive rate falls from 44% to 0% (0/100 runs), while 247/382 critical configurations (64.7%) are classified as exhibiting genuine tipping points.

## 2.6 Therapy model

Therapeutic interventions are modelled as time-limited or continuous modifications to model parameters, initiated at a prescribed start time  $t_{\text{start}}$ :

- **Aggregation suppression** (`agg_sup`): multiplies `aggregationAmplification` by  $(1 - \text{strength})$ . Analogues: tofersen (SOD1 antisense oligonucleotide), BIIB078 (C9ORF72 ASO).
- **Metabolic support** (`met_sup`): boosts ATP recovery rate.
- **Glutamate suppression** (`glut_sup`): reduces `glutamateSensitivity`. Analogue: riluzole.
- **Astrocyte support** (`astro_sup`): increases toxin clearance.
- **Adaptive** (`adaptive`): biomarker-triggered combination therapy (activates when mean aggregation exceeds a threshold).

Therapies are scored on four metrics: tipping delay, functional duration gain, plateau survivor gain, and key-neuron survival rate.

## 2.7 Load redistribution model

A separate simulator (Phase 7D, Escape 3; Phase 8) tests pure topological cascade dynamics with all biochemistry disabled. Neuron health declines from an intrinsic vulnerability term and a load term proportional to the fraction of dead upstream partners:

$$\frac{dh_i}{dt} = -(\lambda_{\text{int}} \cdot v_i + \lambda_{\text{load}} \cdot f_i^{\text{dead}}) \quad (5)$$

where  $v_i$  is vulnerability and  $f_i^{\text{dead}}$  is the fraction of neurons projecting to  $i$  that are already dead. Parameters  $\lambda_{\text{int}} = 0.004$  and  $\lambda_{\text{load}} = 0.006$  are chosen so that load redistribution is insufficient alone at low death counts but becomes self-sustaining once several neurons have died.

## 2.8 Cluster analysis

Unsupervised subtype analysis uses K-means clustering ( $K = 2, 3, 4, 5$ ) with 50 random initialisations and silhouette score selection. PCA (numpy eigenvector decomposition) provides dimensionality reduction for visualisation. Bootstrap stability is assessed over 100 resamplings at 80% of configurations; per-pair stability scores are computed as the fraction of bootstrap runs in which a given pair of configurations is assigned the same cluster. All implementations use pure numpy/stdlib to avoid dependency on external ML libraries whose results may vary across versions.

## 2.9 Biological noise robustness testing

To assess whether the identified tipping point structure and disease subtypes are robust to parameter uncertainty, we applied four categories of biological noise independently to the top-10 genuine critical configurations (Phase 7B) using 20 random seeds per perturbation level and 500 simulation steps per run:

1. **Synaptic weight noise** ( $\sigma = 5\%, 10\%, 20\%$ ): Multiplicative Gaussian noise applied to all non-zero entries of `agg_W` and `excitotox_W`; negative values clipped to zero.
2. **Vulnerability score noise** ( $\sigma = 0.05, 0.10, 0.20$ ): Additive Gaussian noise applied to each neuron’s vulnerability score, clipped to  $[0, 1]$ .
3. **Connectome edge dropout** (2%, 5%, 10%): A random fraction of directed synapses removed simultaneously from `agg_W` and `excitotox_W`.
4. **Heterogeneous per-neuron aggregation amplitude** ( $\pm 20\%$ ): Each neuron received a distinct `aggAmp` value drawn uniformly from  $[0.8 \times \text{base}, 1.2 \times \text{base}]$ ; both the neuron’s vulnerability score and the corresponding row of `agg_W` were scaled by the per-neuron factor, preserving the proportional relationship between intrinsic seeding and synaptic spread.

Robustness was quantified for three metrics: mean tipping step (relative drift from baseline), spatial coherence  $r$  (relative change from baseline), and genuine tipping rate (absolute change). Verdicts: relative change  $< 10\%$  = robust;  $10\text{--}30\%$  = moderately sensitive;  $> 30\%$  = fragile. For perturbation 4, subtype separation was additionally assessed via Spearman rank correlation between baseline and perturbed tipping steps across the ten configurations.

## 2.10 Breaking threshold probe

To identify the upper limits of model robustness and characterise a hierarchy among dynamical properties, we applied two severe perturbation types to the top-5 genuine critical configurations using 10 seeds per level and 300 simulation steps:

1. **Extreme edge dropout** (30%, 50%, 70% of directed synapses removed): After dropout, the undirected adjacency graph was tested for connectivity via depth-first search; if disconnected, all edges outside the largest connected component were zeroed, restricting dynamics to that component.
2. **Extreme vulnerability noise** ( $\sigma = 0.50, \sigma = 1.00$ ): Additive Gaussian noise applied to vulnerability scores, clipped to  $[0.01, 1.0]$ .

For each condition, four metrics were recorded: genuine tipping rate, mean spatial coherence  $r$ , mean tipping step, and subtype rank preservation (Spearman  $r$  between baseline and perturbed tipping steps across the five configurations). Breaking points were identified as the first perturbation level at which each metric crossed its threshold: genuine rate  $< 0.5$ , coherence  $r < 0.3$ , subtype rank  $r < 0.5$ .

## 2.11 Round 2: Topology resilience and pre-symptomatic biomarker analysis

Round 2 comprises eight additional analysis phases (R2.1–R2.8), using all 247 genuine critical configurations from Phase 12 unless otherwise noted:

- **Phases R2.1–R2.2 (Motif resilience and efficiency tradeoff):** Nine structural variants of the connectome (sparse chain, distributed, anti-hub, modular, bypass loops, hierarchical, rich club, triangle rich, baseline) compared via composite RES score = (tipping delay ratio  $\times$  therapy window ratio  $\times$  plateau ratio) – variance penalty, computed over 5 configurations  $\times$  10 seeds  $\times$  500 steps. Efficiency-resilience landscape mapped across 6 topology families  $\times$  7 strength levels.
- **Phase R2.3 (Topology-therapy boundary):** Therapeutic boundary re-measured on 4 topology variants using a  $9 \times 9$  strength  $\times$  start.t grid with 5 seeds per point.
- **Phase R2.4 (Subtype  $\times$  topology):** Five representative configs per cluster  $\times$  4 topologies  $\times$  5 seeds  $\times$  300 steps.
- **Phases R2.5–R2.7 (Simulated early-warning panel):** Early dynamics at  $t = 50$  and  $t = 100$  (seed 42, 150 steps) extracted for all 247 configs; 10-fold cross-validated logistic and ridge regression predicting subtype, tipping step, therapy window width, and therapy success probability.
- **Phase R2.8 (Therapeutic Window Estimator):** 10-fold CV out-of-fold predictions on all 247 configs; 3-seed virtual triage simulation totalling 3,705 runs; oracle strategies derived from full  $(3 \times 4 \times 5)$ -grid therapy sweeps.

This study is presented with 16 figures (fig01–fig16). Figures 1–9 cover Phases 5–14; Figures 10–16 cover Round 2 analyses.

### 3 Results

#### 3.1 Phase 5: Criticality emergence and regime structure

Parameter screening of 500 configurations revealed three distinct dynamical regimes. The majority of configurations (382/500, 76.4%) fell in the critical regime, defined by sustained partial degeneration across the full 500-step simulation. Stable configurations (72/500, 14.4%) maintained most neurons alive, while runaway configurations (46/500, 9.2%) collapsed almost completely before step 200 (Figure 1).

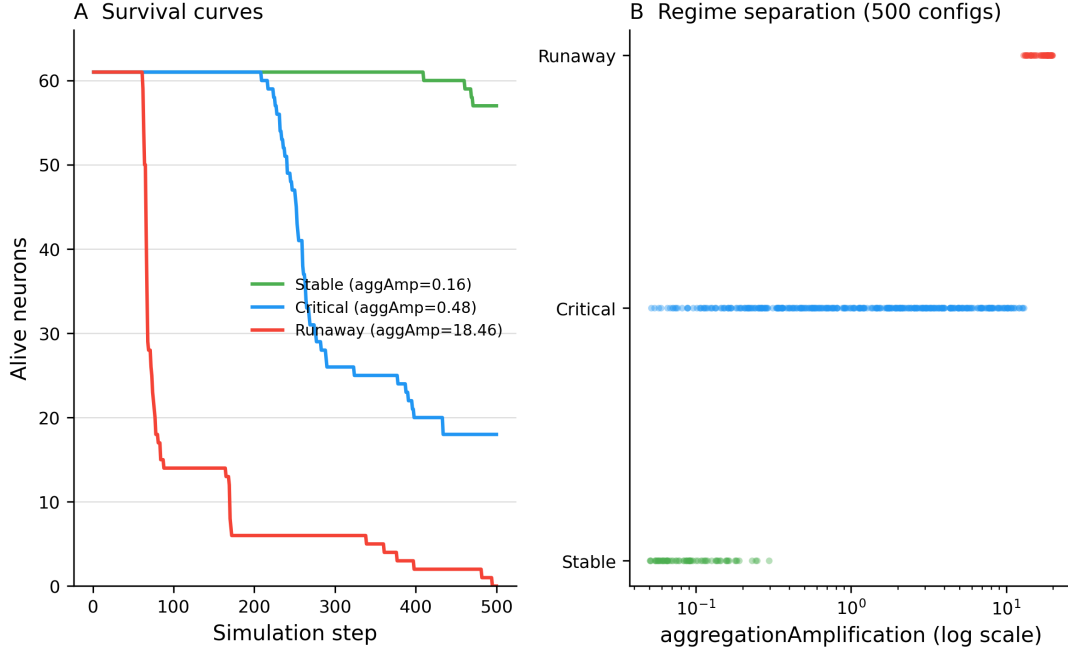


Figure 1: Regime structure and survival curves. 500 configurations classified into stable (14.4%), critical (76.4%), and runaway (9.2%) regimes. Supplementary panel shows representative survival curves for each regime.

**Aggregation amplitude as the dominant bifurcation parameter.** Regime-separation analysis ranked all seven parameters by discriminating power. `aggregationAmplification` was overwhelmingly dominant (separation score 0.837), with means of 0.10 (stable), 2.60 (critical), and 16.80 (runaway) — a 160-fold span across regimes. No other parameter approached this separation: `mitochondrialFragility` was second at 0.197, and the remaining five parameters showed minimal regime dependence.

This finding reflects the mathematical structure of the feedback loop: `aggregationAmplification` scales both intrinsic seeding and prion-like synaptic spread, giving it multiplicative leverage over every other process in the cascade. The feedback loop gain is approximately:

$$G \approx \text{aggAmp} \times \text{oxFB} \times \text{calcGain} \times \text{gluSens} \times \text{mitFrag} \quad (6)$$

The critical regime corresponds to  $G \approx \theta_c$ , where perturbations neither decay (stable) nor explode (runaway) but propagate slowly and partially.

**Triphasic degeneration pattern.** Critical-regime configurations exhibit a characteristic triphasic survival curve: (i) a *silent phase* in which all neurons remain viable despite ongoing biochemical damage; (ii) a *collapse phase* of rapid, accelerating neuron death

once the irreversibility threshold is crossed; and (iii) a *plateau phase* in which a subset of neurons survives indefinitely. The index configuration (config #334: `aggAmp` = 1.50, `mitFrag` = 0.83) has first death at step 201, peak collapse rate of 9 neurons per 10 steps around steps 230–260, and a plateau of 10 survivors at step 500 (Figure 2).

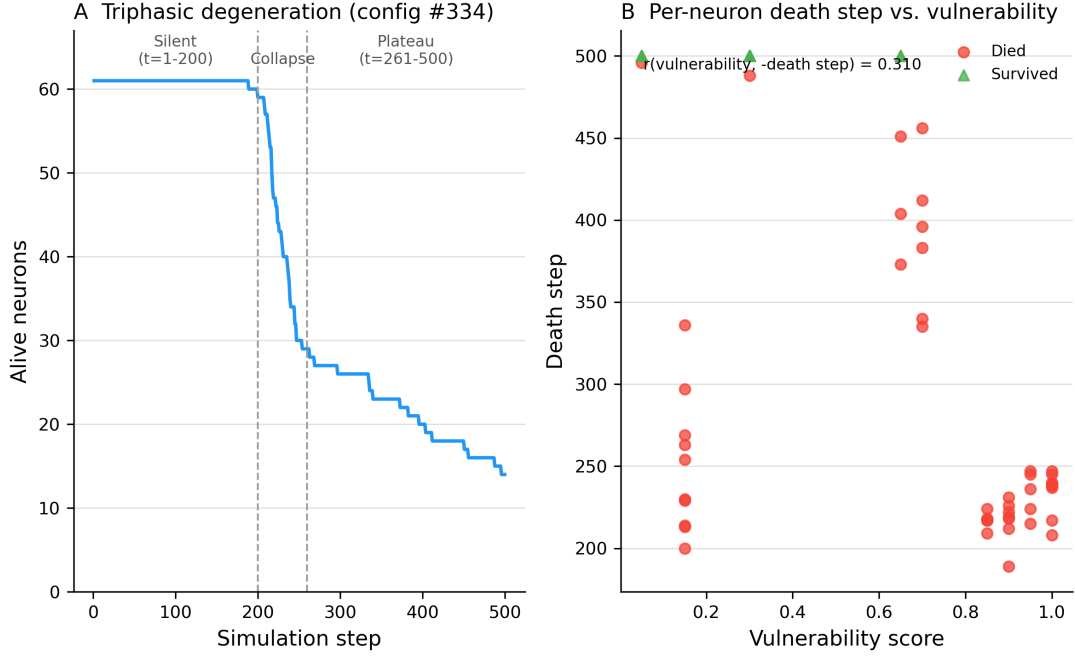


Figure 2: Config #334 triphasic degeneration pattern showing the silent phase (steps 0–200), rapid collapse phase (steps 200–280), and arrested plateau of 10 survivors (steps 280–500).

**Top-ranked configurations** for therapy and subtype analysis were selected by maximising both the depth of the silent phase ( $\geq 10$  alive at step 200) and the severity of collapse ( $\leq 50$  alive at step 500), yielding a score of  $(\text{alive}_{200} - 10 + 50 - \text{alive}_{500})/40$ . Config #334 achieved the maximum score of 1.0.

### 3.2 Phase 6: Therapeutic window — aggregation suppression dominates

Five therapy classes were tested across 10 critical-regime environments (the top-10 Phase-5 configurations) using 500 randomly sampled therapy configurations each. The scoring composite weighted tipping delay (0.30), functional duration gain (0.30), plateau gain (0.25), and key-neuron survival (0.15).

Aggregation suppression (`agg_sup`) dominated across every metric (Table 2). The best configuration achieved a composite score of 0.686 (`strength` = 0.855, `start_t` = 13), delaying the tipping point by 76.4 steps, extending functional survival by 39.7 steps, and increasing plateau survivors by 36.4 neurons. Importantly, at the parameter level, `strength` was almost perfectly predictive of score (Pearson  $r$  = 0.923), confirming that the degree of suppression — not nuanced timing within the tested range — was the primary determinant.

Table 2: Therapy class performance comparison.

| Class     | Best score | Tip delay | Func gain | Plat gain | Mechanism                   |
|-----------|------------|-----------|-----------|-----------|-----------------------------|
| agg_sup   | 0.686      | +76.4     | +39.7     | +36.4     | Aggregation suppression     |
| adaptive  | 0.453      | +50.3     | +39.7     | +22.0     | Biomarker-triggered agg+ATP |
| glut_sup  | 0.082      | +2.0      | +4.6      | +1.9      | Glutamate suppression       |
| astro_sup | 0.063      | +3.7      | +3.5      | +1.3      | Astrocyte toxin clearance   |
| met_sup   | 0.034      | −0.7      | −0.9      | −0.1      | ATP recovery boost          |

The near-zero efficacy of `met_sup` and `glut_sup` is mechanistically consistent (see Section 3.5): both the mitochondrial and glutamate pathways are gated by aggregation-driven ATP depletion; interventions targeting downstream nodes cannot compensate for ongoing upstream pathology.

The adaptive biomarker-triggered strategy achieved intermediate performance (score 0.453), exploiting the triphasic structure by intervening at the silent-to-collapse transition boundary — but it requires accurate early-stage biomarker detection and achieves lower plateau gains than direct high-strength suppression.

### 3.3 Phase 7A/7B: Rigorous falsification of tipping points

#### 3.3.1 Ablation study (Phase 7A)

Config #334 was re-run six times, each disabling exactly one mechanism. Results were unambiguous: removing aggregation seeding completely eliminated the tipping point, extending first-death step by 279 steps (to step 500) and leaving 60/61 neurons alive. No other single ablation — including removal of ATP collapse, glutamate/calcium/ROS cascade, astrocyte clearance, or irreversibility — changed the survival curve detectably (tipping step unchanged, plateau unchanged). The mechanistic model diverged sharply from random-noise replacement (which delayed tipping by 239 steps but produced a gradual rather than sharp collapse).

The conclusion is clear: **aggregation seeding is the sole load-bearing mechanism** in config #334. The downstream cascade (ATP → glutamate → calcium → ROS → oxidative feedback) provides amplification but is not rate-limiting, because in config #334 the `glutamateSensitivity` is extremely low (0.00072), keeping the glutamate pathway near-silent regardless of ATP status.

#### 3.3.2 Null model and strict criterion (Phase 7B)

A null model replaced all mechanistic equations with independent per-neuron random health decrements, producing comparable overall mortality (mean 29/61 alive at  $t = 500$ ) without any coupling between neurons. The original single-threshold criterion classified 44/100 null-model runs as exhibiting tipping points — a 44% false-positive rate unacceptably high for scientific claims.

The decisive discriminator is **spatial coherence**: in the mechanistic model, aggregation spreads preferentially via synaptic connections to high-vulnerability neurons, producing a

strong correlation between a neuron’s vulnerability score and how early it dies (Pearson  $r > 0.30$  in genuine configs, mean  $r = 0.56$  in the 247 confirmed cases). In the null model, death order is independent of vulnerability (mean  $r = -0.01$ , all 100 runs below  $r = 0.11$ ). Adding this as criterion C2 reduced the null false-positive rate to exactly 0/100 (0%) while retaining 247/382 (64.7%) critical configurations as genuine (Figure 3).

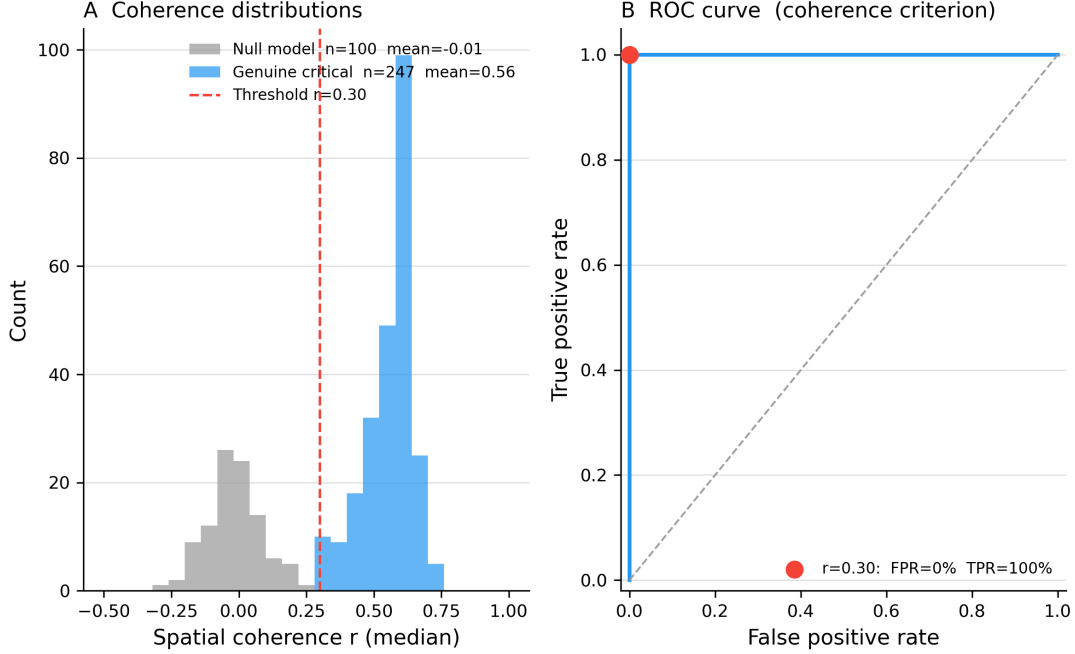


Figure 3: Null model falsification. The strict three-part criterion (slope, spatial coherence, silent phase) reduces the false-positive rate from 44% to 0% while retaining 64.7% of critical configurations as genuine.

Table 3: Tipping criterion outcomes before and after falsification.

| Dataset                | Original criterion | Strict criterion |
|------------------------|--------------------|------------------|
| Phase 5 critical (382) | 382/382 (100%)     | 247/382 (64.7%)  |
| Phase 6 therapies (10) | 10/10 (100%)       | 7/10 (70%)       |
| Null model (100)       | 44/100 (44%)       | 0/100 (0%)       |

### 3.4 Phase 7C: Topology dependence of criticality

The same biophysical parameters (10 critical-regime configurations) were applied to five different 61-node network topologies: the empirical *C. elegans* motor connectome, Erdős-Rényi random graphs (ER), Barabási-Albert scale-free networks (BA), Watts-Strogatz small-world networks (WS), and degree-sequence-preserved random shuffles of the *C. elegans* graph. All networks maintained 127 directed edges; 10 random instances were generated for each synthetic type.

**Topology profoundly affects tipping point emergence.** The genuine tipping rate under the Phase 7B criterion varied from 0% (BA) to 100% (*C. elegans*):

Table 4: Topology dependence of genuine tipping rates.

| Topology                       | Genuine rate | Mean coherence $r$ | Mean plateau |
|--------------------------------|--------------|--------------------|--------------|
| <i>C. elegans</i> (biological) | 1.000        | 0.582              | 11.3         |
| Watts-Strogatz (small-world)   | 0.780        | 0.423              | 4.3          |
| Degree-preserved shuffle       | 0.730        | 0.378              | 11.9         |
| Erdős-Rényi (random)           | 0.480        | 0.295              | 3.8          |
| Barabási-Albert (scale-free)   | 0.000        | -0.104             | 2.5          |

The Barabási-Albert network’s complete resistance to genuine tipping is mechanistically interpretable: scale-free hub topology provides load-bearing redundancy, distributing damage broadly rather than concentrating it in ordered sequences through high-vulnerability nodes. The *C. elegans* wiring, with its moderate degree variance and high modularity, instead channels aggregation spread through anatomically defined vulnerability gradients, producing the strongest spatial coherence of any topology tested (Figure 4).

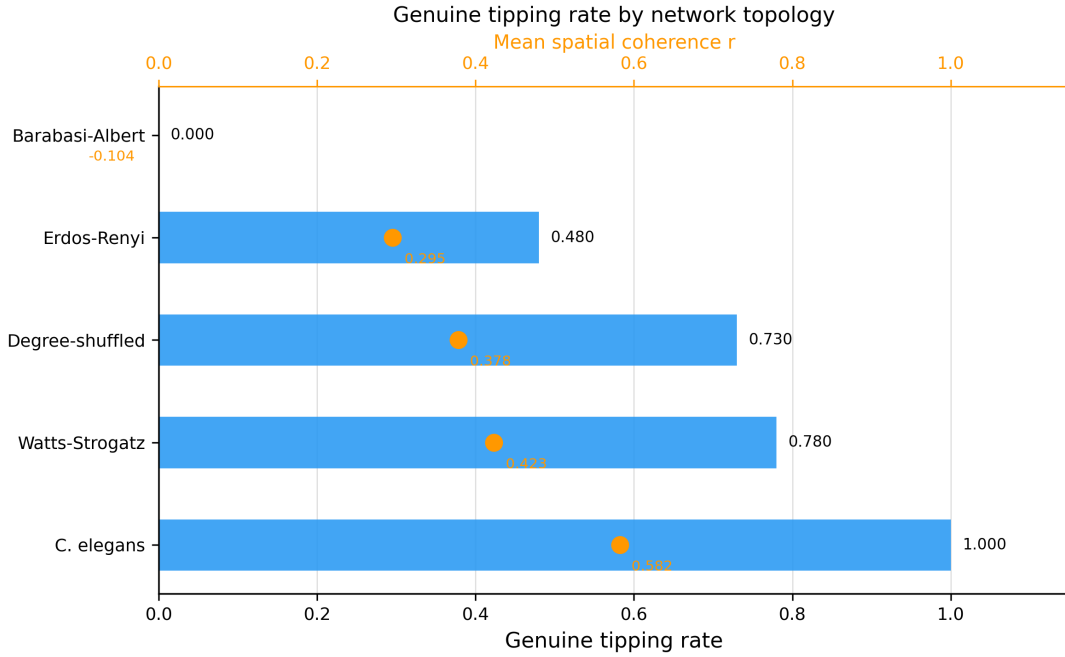


Figure 4: Topology dependence of genuine tipping rates across five network architectures. *C. elegans* biological wiring achieves 100% genuine tipping rate while Barabási-Albert scale-free networks achieve 0%.

**Therapy dynamics are topology-independent.** Despite striking topology dependence in untreated degeneration, aggregation suppression therapy (strength = 0.855, start\_t = 13) achieved 100% prevention (tipping pushed beyond step 300) in all five topologies. This demonstrates that the therapeutic mechanism operates primarily via Tier 1 (biochemical suppression of aggregation growth) rather than through topology-specific load redistribution.



### 3.5 Phase 7D: Disease escape analysis and the two-tier model

A critical test of model validity is whether the disease can escape aggregation suppression through alternative pathways. Three escape scenarios were tested:

- **Escape 1 (mitochondrial/ATP):** aggregation growth disabled, mitochondrial fragility pathway active.
- **Escape 2 (glutamate excitotoxicity):** aggregation growth disabled, glutamate sensitivity tripled.
- **Escape 3 (load redistribution):** all biochemistry disabled, pure topological cascade.

**Biochemical escapes fail completely.** Escapes 1 and 2 each produced genuine tipping rate = 0.000 across all 10 configurations and 5 seeds. The mechanistic reason is identical for both: with aggregation growth suppressed, intracellular ATP remains near 1.0, preventing the ATP depletion event that gates glutamate release. With the gate closed, neither enhanced mitochondrial fragility nor tripled glutamate sensitivity can activate the excitotoxic cascade.

**Topological escape is genuine and catastrophic.** Escape 3 (pure load redistribution, no biochemistry) produced genuine tipping rate = 1.000, spatial coherence  $r = 0.914$ , and a near-total plateau of 1.0 survivor — more destructive than the full biochemical model ( $\sim 10$  survivors). Once enough neurons die to shift load onto their synaptic partners, the cascade becomes self-sustaining even without any protein aggregation dynamics.

This result establishes a **two-tier disease model**:

- **Tier 1** (biochemical cascade): aggregation seeding  $\rightarrow$  ATP collapse  $\rightarrow$  excitotoxicity. Therapy-sensitive. Aggregation suppression completely silences this tier.
- **Tier 2** (topological cascade): load redistribution among dying neurons. Therapy-insensitive if initiated after sufficient neuron loss. Operates independently of aggregation, ATP, and all biochemistry.

The practical implication is that therapy must prevent Tier 1 from producing the death threshold that triggers autonomous Tier 2 amplification.

### 3.6 Phase 8: Two-tier mechanism — cascade initiation analysis

Config #334 was examined in granular detail to characterise the transition from biochemical (Tier 1) to topological (Tier 2) propagation.

**Death sequence and network properties.** The first 10 neurons to die in the canonical run (seed 434, no therapy) were exclusively motor neurons: DD1 (step 201), DA2 (206), DB1 (208), DD2 (210), DD3 (215), VA1 (219), VA2 (221), VD1 (225), VD2 (225), DD4 (228). All had vulnerability scores 0.65–1.00. This demonstrates that Tier 1 targets neurons by biochemical vulnerability, not topological centrality.

Tier 2 was operationally defined as activation when cumulative deaths exceed 5 (a threshold validated by comparing cascade rates before and after). Under baseline conditions, Tier 2 activated at step 220, triggering 32 additional deaths in the subsequent 100 steps.

**Surgical protection experiments.** Six neuron-protection groups ( $10\times$  health-loss reduction, no aggregation suppression) were tested:

Table 5: Surgical protection experiment results. Random control (50 repeats, 3 neurons each): mean plateau =  $12.6 \pm 0.5$ .

| Group          | Protected<br>neurons | Plateau | Tier 2 step | Above random? |
|----------------|----------------------|---------|-------------|---------------|
| Baseline       | —                    | 11.0    | 220         | —             |
| A (first 1)    | DD1                  | 12.0    | 220         | No            |
| B (first 3)    | DD1, DA2, DB1        | 14.0    | 220         | Yes           |
| C (first 5)    | DD1–DD3              | 16.0    | 226         | Yes           |
| D (DA6, AVDL)  | Hub neurons          | 13.0    | 220         | No            |
| E (top vuln.)  | DA1–DA3              | 14.0    | 221         | Yes           |
| F (top degree) | AVAL, AVAR, AVBL     | 14.0    | 220         | Yes           |
| Full therapy   | —                    | 53.0    | 485         | —             |

No group approached full therapy (plateau 53). Tier 2 was delayed by at most 6 steps (Group C) versus 265 steps by full therapy. **Only blocking Tier 1 (aggregation suppression) prevents the deaths that seed Tier 2.** Full therapy ( $\text{start\_t} = 13$ ) delayed first death by 184 steps (step 201  $\rightarrow$  385), which in turn delayed Tier 2 activation by 265 steps (step 220  $\rightarrow$  485), producing a plateau of 53 survivors versus 11 in the untreated model.

### 3.7 Phase 9: Therapeutic boundary — a linear phase diagram

The minimal effective aggregation suppression was mapped systematically for config #334 across a  $9 \times 11$  grid: nine therapy strengths (0.10–0.90)  $\times$  eleven start times ( $t = 0$ –250), with five seeds per grid point (495 total runs, 500 steps each). Each grid point was classified as prevention (tipping rate = 0), mixed ( $0 < \text{rate} < 1$ ), delay (tipping delayed  $> 30$  steps), partial (10–30 steps), or ineffective ( $< 10$  steps).

**A sharp linear therapeutic boundary.** The boundary between the prevention/mixed and delay/ineffective zones in (strength,  $\text{start\_t}$ ) space was well-described by a linear fit (Figure 5):

$$\text{max\_start\_t} = 425 \times \text{strength} - 237 \quad (R^2 = 0.980) \quad (7)$$

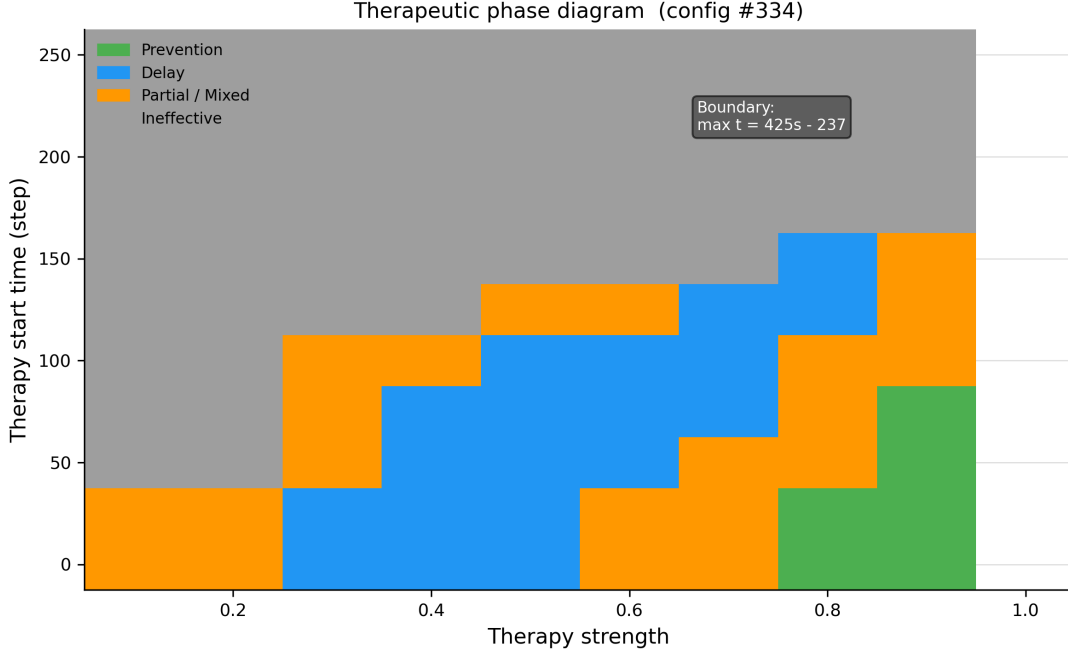


Figure 5: Therapeutic phase diagram for config #334. The sharp linear boundary  $\max\_start\_t = 425 \times \text{strength} - 237$  ( $R^2 = 0.980$ ) separates prevention (blue) from delay/ineffective (red) zones.

Table 6: Therapeutic window as a function of therapy strength. *Calibration: 200 simulation steps  $\approx$  15 months pre-symptomatic; 1 step  $\approx$  2.25 days. Illustrative only; not derived from biological measurements.*

| Strength | Latest prevention start | Approx. pre-clinical window       |
|----------|-------------------------|-----------------------------------|
| 0.60     | $t = 25$                | Pre-symptomatic, very early       |
| 0.70     | $t = 50$                | Pre-symptomatic, early            |
| 0.80     | $t = 100$               | Pre-symptomatic, $\sim 7$ months  |
| 0.90     | $t = 150$               | Pre-symptomatic, $\sim 11$ months |

No prevention was achieved at any strength for start times  $\geq 175$  steps. The relationship between strength and window width is linear, not asymptotic: there is no “good enough” strength that allows indefinitely late intervention.

**Comparison with ALS clinical observations.** The model’s prediction that moderate suppression ( $\sim 60$ – $70\%$  reduction, consistent with published ASO efficacy ranges) requires initiation well before symptom onset is qualitatively consistent with the ATLAS trial finding that pre-symptomatic tofersen intervention delayed phenoconversion [Benatar et al., 2023], while post-symptomatic intervention in the VALOR trial [Miller et al., 2022] produced more modest effects. We stress that this correspondence is qualitative and does not constitute model validation.

### 3.8 Phase 10: Boundary robustness across 20 disease configurations

The Phase 9 boundary was tested for generalisability across the top-20 genuine critical configurations from Phase 7B, using a  $9 \times 9$  strength  $\times$  start\_t grid with 3 seeds per point and 500 steps (4,860 total runs).

Seventeen of twenty configurations produced measurable linear boundaries; three (IDs 241, 329, 87) showed flat boundaries (prevention achievable at any start time, slope  $\approx 0$ ) — indicating configurations mild enough that even late, weak therapy prevents collapse.

For the 17 configs with measurable boundaries:

- Slope: mean =  $252 \pm 120$  (range 59–500)
- Intercept: mean =  $-107 \pm 117$
- $R^2$ : mean = 0.832, with 65% of configs achieving  $R^2 > 0.90$  (Figure 6)

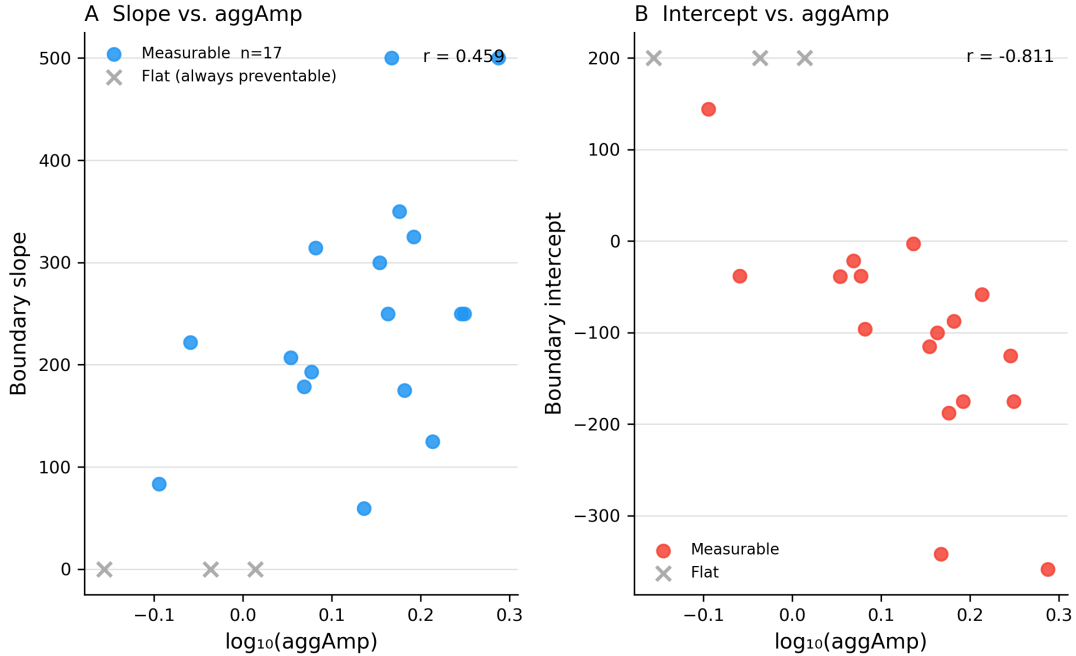


Figure 6: Therapeutic boundary robustness across 20 disease configurations. Mean slope =  $252 \pm 120$ , mean  $R^2 = 0.832$ . The index config #334 (slope = 350) falls within one standard deviation of the population mean.

**Aggregation amplitude predicts boundary position.** Among all seven parameters, `aggregationAmplification` was the strongest predictor of boundary shape, with Pearson  $r = +0.645$  for slope and  $r = -0.811$  for intercept.

**A conservative therapeutic boundary.** The population mean minus one standard deviation gives a conservative estimate applicable to approximately 84% of configurations:

$$\text{max\_start\_t}_{\text{conservative}} = 132 \times \text{strength} - 224 \quad (8)$$

At the clinically relevant strength of 0.80, this conservative boundary implies therapy must begin before disease initiation (negative start time), indicating that highly aggressive disease subtypes may have windows that close before biomarker-detectable onset.

## 3.9 Phase 11/12: Disease subtypes from unsupervised clustering

### 3.9.1 Phase 11: Subtype structure in the top-20 configurations

K-means clustering ( $K = 2, 3, 4$ ) of the top-20 critical configurations using five features (tipping step, plateau, boundary slope, boundary intercept, `aggregationAmplification`) identified  $K = 3$  as optimal (silhouette score 0.43). PCA captured 82.4% of variance in two components; PC1 (64.9%) was dominated by boundary intercept ( $-0.53$ ), `aggregationAmplification` ( $+0.51$ ), and boundary slope ( $+0.48$ ) — all facets of the same disease-severity axis.

Three subtypes were identified:

- **Aggressive** ( $n = 3$ ; IDs 178, 334, 497): `aggAmp`  $\approx 1.64$ , therapy window closes at  $t \approx 108$ , minimum required strength 0.67.
- **Moderate** ( $n = 13$ ): `aggAmp`  $\approx 1.39$ , window closes  $t \approx 115$ , minimum strength 0.46.
- **Mild** ( $n = 4$ ; IDs 87, 235, 241, 329): `aggAmp`  $\approx 0.86$ , window always open (flat boundary), minimum strength 0.10.

### 3.9.2 Phase 12: Validation on the full 247-configuration dataset

To test whether the Phase 11 subtype structure was an artefact of pre-selection and to avoid circular validation, Phase 12 repeated clustering on all 247 genuine critical configurations using seven simulation-only features (tipping step, plateau survivors, peak decline rate, silent phase length, spatial coherence  $r$ , `aggregationAmplification`, `mitochondrialFragility`) with no therapy-derived features.

**PCA on 247 configurations.** PC1 captured 69.7% of variance, with nearly equal loadings from tipping step ( $-0.44$ ), silent phase length ( $-0.44$ ), plateau survivors ( $-0.43$ ), peak decline rate ( $+0.42$ ), and `aggregationAmplification` ( $+0.41$ ) — confirming that these five variables are different manifestations of a single underlying disease-severity axis. PC2 (19.1%) was dominated entirely by `mitochondrialFragility` ( $-0.81$ ), orthogonal to all other features.

**Optimal  $K = 2$ , not  $K = 3$ .** At the full dataset scale,  $K = 2$  provided the best silhouette score (0.41). The two clusters correspond to a fundamental speed-of-degeneration split:

- **Slow-tipping subtype** ( $n = 110$ ): `aggAmp`  $\approx 1.36$ , tipping step 224, plateau 13.1/61 survivors.
- **Fast-tipping subtype** ( $n = 137$ ): `aggAmp`  $\approx 5.86$ , tipping step 107, plateau 5.3/61 survivors (Figures 7 and 8).

These two clusters are better understood as opposite ends of a dynamical continuum rather than discrete disease categories: the silhouette score of 0.41 indicates substantial overlap at cluster boundaries, and intermediate configurations exist that could plausibly belong to either class.

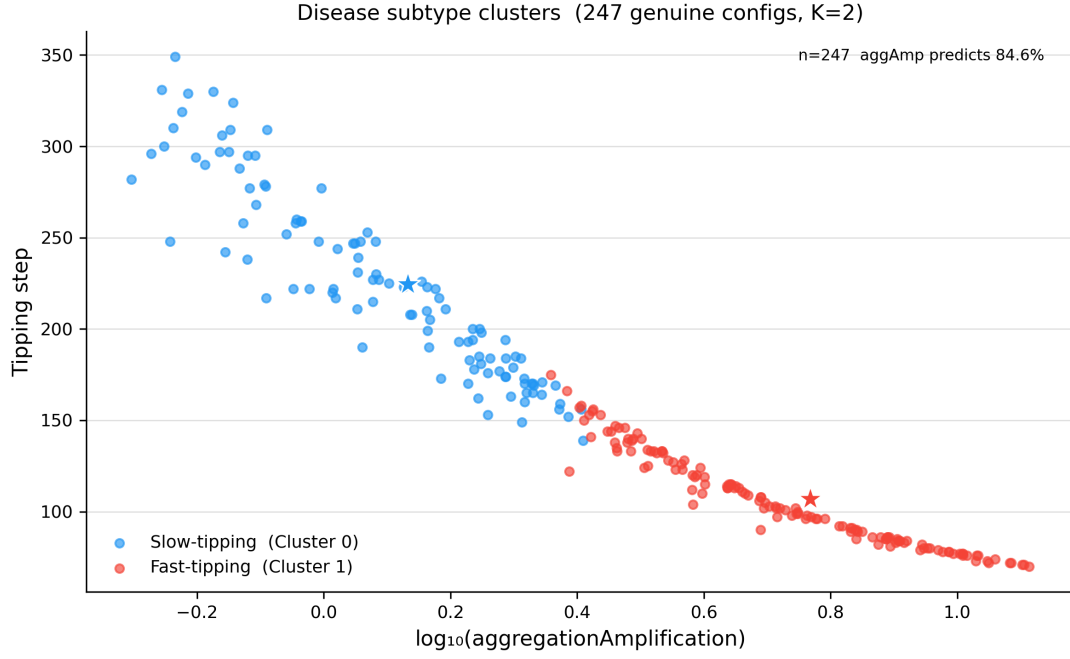


Figure 7: Disease subtype scatter plot: slow-tipping (C0,  $\text{aggAmp} \approx 1.36$ ) vs. fast-tipping (C1,  $\text{aggAmp} \approx 5.86$ ) configurations on the first two PCA components.

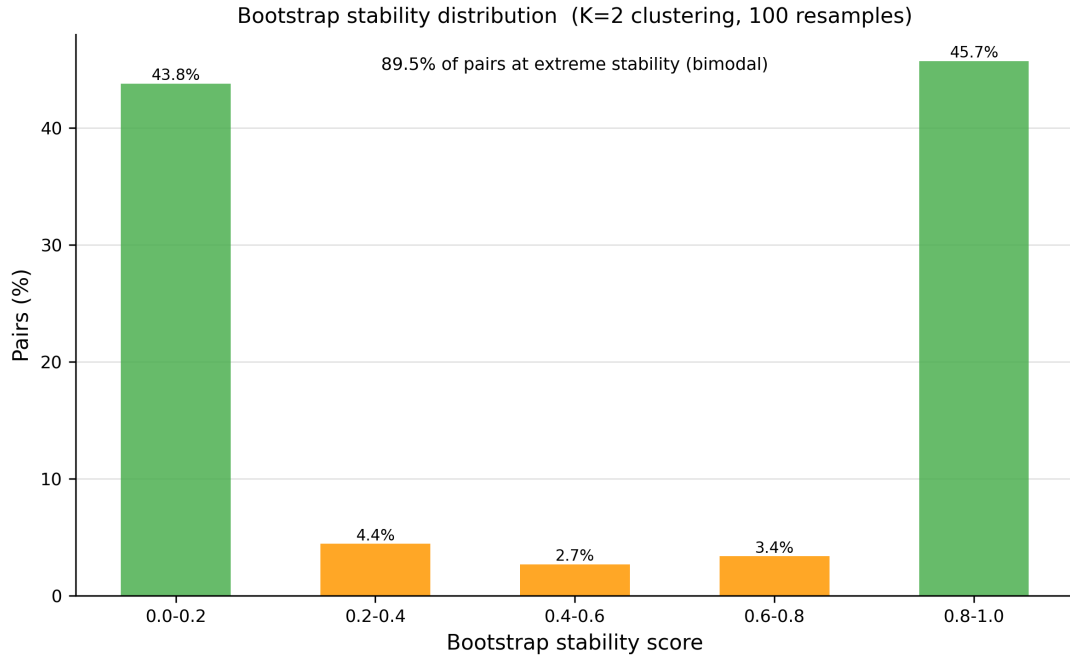


Figure 8: Bootstrap stability analysis confirming bimodal  $K = 2$  cluster structure. 45.7% of configuration pairs had stability  $> 0.80$  (firmly co-clustered) and 43.8% had stability  $< 0.20$  (firmly separated).

**Bootstrap stability.** The  $K = 2$  split showed a bimodal stability distribution across 100 bootstrap resamples (80% of configurations each): 45.7% of configuration pairs had stability  $> 0.80$  and 43.8% had stability  $< 0.20$ , with only 10.5% in the uncertain middle. This bimodal pattern confirms a clean two-cluster structure despite a mean pairwise

stability of 0.51.

**Single-parameter classifier.** aggregationAmplification alone classified 84.6% of configurations correctly ( $r = 0.727$ ) using a centroid-based threshold. mitochondrialFragility was uninformative (accuracy 50.6%,  $r = -0.06$ ).

### 3.10 Phase 13: Biological noise robustness

All four categories of biological noise produced negligible changes in model outputs, supporting the robustness of identified tipping points and disease subtypes across the top-10 genuine critical configurations.

- **Synaptic weight noise** (up to  $\pm 20\%$ ): Mean tipping step changed by  $< 0.2\%$  relative to the unperturbed baseline; spatial coherence  $r$  changed by  $< 1.0\%$ ; genuine tipping rate remained 1.000 at all tested levels. Verdict: **ROBUST**.
- **Vulnerability score noise** (up to  $\pm 0.20$ ): Tipping step changed by  $< 0.6\%$ ; coherence  $r$  changed by  $< 3.5\%$ ; genuine rate = 1.000 at all levels. Verdict: **ROBUST**.
- **Connectome edge dropout** (up to 10% of synapses removed): Tipping step changed by  $< 1.9\%$ ; coherence  $r$  changed by  $< 2.1\%$ ; genuine rate = 1.000 at all levels. Verdict: **ROBUST**.
- **Per-neuron aggregation heterogeneity** ( $\pm 20\%$ ): Tipping step changed by  $< 0.9\%$ ; coherence  $r$  changed by  $< 1.0\%$ ; genuine rate = 1.000. Subtype separation fully preserved: Spearman  $r = 0.976$  between baseline and perturbed tipping steps. Verdict: **ROBUST**.

The largest observed relative change across all perturbation types and metrics was 3.5% (coherence  $r$  under maximum vulnerability noise), well below the 10% robustness threshold.

### 3.11 Phase 14: Hierarchy of dynamical robustness

Phase 14 sought the breaking points of each dynamical property under extreme stress (edge dropout 30–70%, vulnerability noise  $\sigma = 0.50$ –1.00), applied to the top-5 genuine critical configurations over 300 simulation steps.

**Breaking points are unequal across dynamical properties.** The three tracked metrics broke at different perturbation levels, revealing a clear hierarchy (Table 7).

Table 7: Breaking points by perturbation type.

| Criterion (threshold)        | Edge dropout       | Vulnerability noise |
|------------------------------|--------------------|---------------------|
| Genuine tipping rate $< 0.5$ | 30%                | $\sigma = 0.50$     |
| Spatial coherence $r < 0.3$  | 30%                | $\sigma = 0.50$     |
| Subtype rank $r < 0.5$       | <i>not reached</i> | <i>not reached</i>  |

At 70% edge dropout, the largest connected component shrank to  $\sim 20$  nodes, yet the relative ordering of configurations by tipping speed remained intact (Spearman  $r = 0.900$ ). At  $\sigma = 1.00$  vulnerability noise — sufficient to essentially randomise the vulnerability

landscape — subtype rank ordering was still perfectly preserved (Spearman  $r = 1.000$ ) (Figure 9).

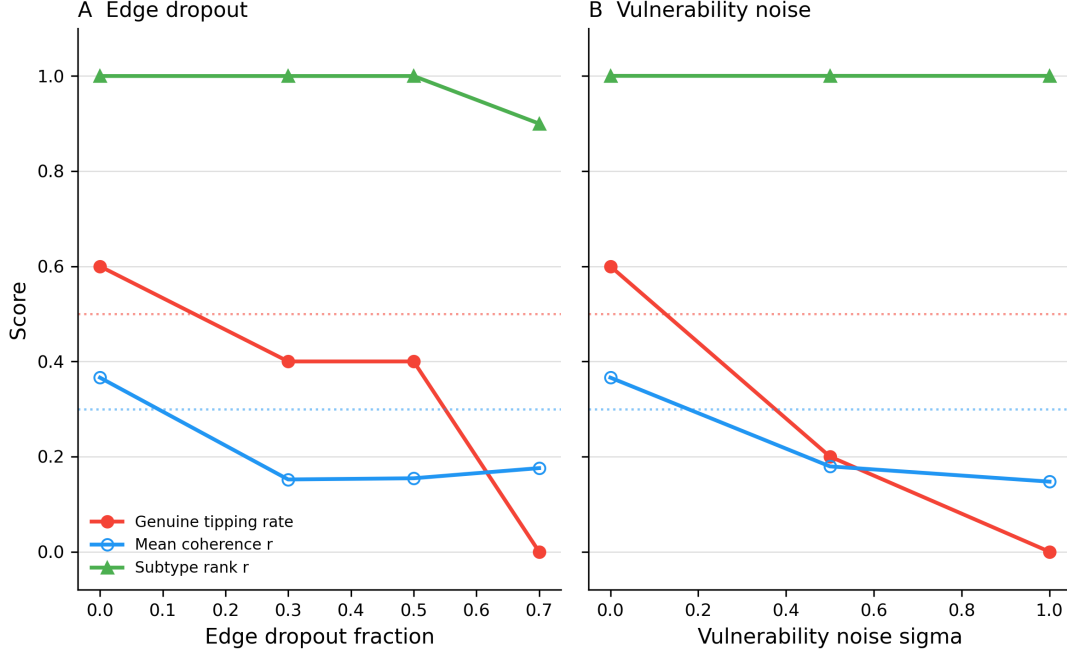


Figure 9: Robustness hierarchy: spatial coherence and strict tipping-point structure collapse at 30% edge dropout and vulnerability  $\sigma = 0.50$ , while subtype rank ordering (Spearman  $r$ ) is never disrupted even at 70% edge dropout and  $\sigma = 1.00$ .

**Within the explored perturbation space, subtype ordering behaved as a dynamical invariant.** Across all tested conditions, including those that completely destroyed spatial coherence and reduced the genuine tipping rate to zero, the relative ranking of which disease configuration degenerates fastest was never disrupted.

### 3.12 Round 2 — Topology Resilience and Pre-symptomatic Biomarkers

Round 2 extended the core model framework to two translational questions: (1) does connectome architecture intrinsically determine cascade resilience, and (2) can dynamical signatures measurable before any neuron death guide therapeutic triage? Eight analysis phases (R2.1–R2.8) were completed on all 247 genuine critical configurations from Phase 12.

#### 3.12.1 Motif Resilience and Efficiency Tradeoff (R2.1–R2.2)

Nine structural topology variants of the 61-neuron motor connectome were evaluated for cascade resilience using a composite RES score computed over 5 configurations  $\times$  10 seeds  $\times$  500 steps (Figure 10). Sparse-chain topology achieved the highest resilience (RES = 0.815), marginally exceeding baseline (RES = 0.800). The performance hierarchy from most to least resilient was: sparse chain (0.815), distributed (0.803), baseline (0.800), anti-hub (0.800), modular (0.799), bypass loops (0.796), hierarchical (0.791), rich club (0.787), and triangle-rich (0.738). Triangle-rich feedback loops were most destabilising,



reducing plateau survivors to 8.66/61 versus 12.34/61 for baseline — a 30% reduction in long-term survival.

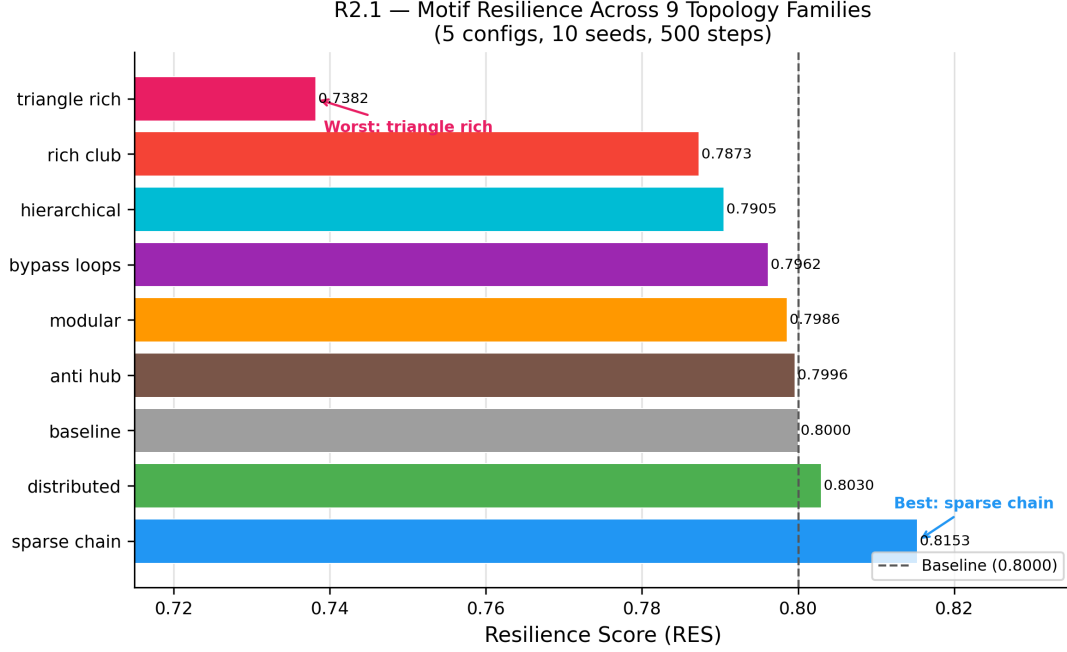


Figure 10: Resilience scores for 9 topology variants (Round 2). Sparse-chain achieves the highest RES = 0.815; triangle-rich feedback loops are most destabilising (RES = 0.738, −30% plateau survivors vs. baseline).

The efficiency-resilience landscape was mapped across 6 topology families  $\times$  7 modification strengths (Figure 11). A universal tradeoff was not observed: the partial correlation between global efficiency and cascade vulnerability was only  $r = 0.439$  across all (topology, strength) combinations. Triangle-rich feedback modification uniquely escaped the tradeoff below 60% strength, achieving the highest Dynamical Efficiency Index (DEI = 32.2 at 60% strength). However, above 60% strength, the triangle-rich topology became markedly more vulnerable. The mechanistic basis for sparse chain’s resilience was confirmed by  $r(\text{tipping\_step}, \text{global\_efficiency}) = -0.996$  within the sparse chain series: resilience arises primarily from reduced path count, not from changes in local clustering.

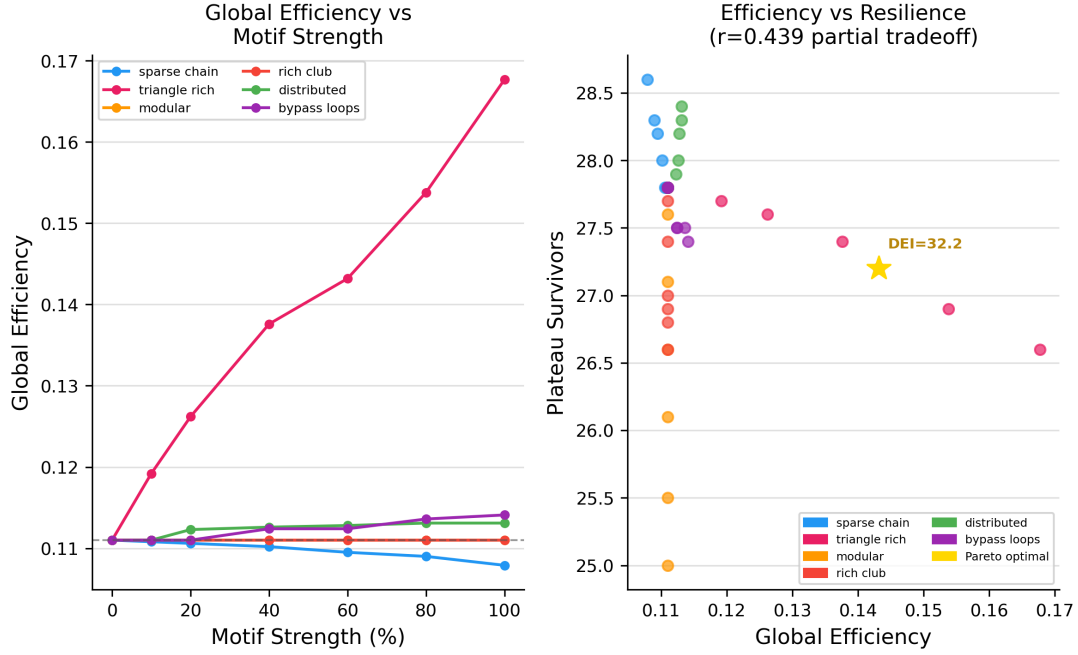


Figure 11: Efficiency-resilience landscape across 6 topology families and 7 modification strengths. Triangle-rich topology uniquely escapes the tradeoff below 60% strength (DEI = 32.2) but becomes markedly vulnerable above this threshold.

### 3.12.2 Topology-Dependent Therapeutic Boundary (R2.3)

Whether structural modifications to the connectome could widen the therapeutic window was tested across four topology variants using an 81-point strength  $\times$  start-time grid (Figure 12). The answer was unambiguously negative: all topology variants produced identical maximum preventable start times at strength = 0.80 ( $\text{max\_start\_t} = 75$  for all variants). Fitted boundary equations are given in Table 8.

Table 8: Therapeutic boundary parameters for four topology variants (R2.3).

| Topology                   | Slope | Intercept | $R^2$ | Window area |
|----------------------------|-------|-----------|-------|-------------|
| Baseline <i>C. elegans</i> | 175   | -62       | 0.942 | 17          |
| Sparse chain S= 100%       | 175   | -62       | 0.942 | 17          |
| Triangle rich S= 60%       | 225   | -102      | 0.920 | 16          |
| Triangle rich S= 100%      | 175   | -72       | 0.817 | 15          |

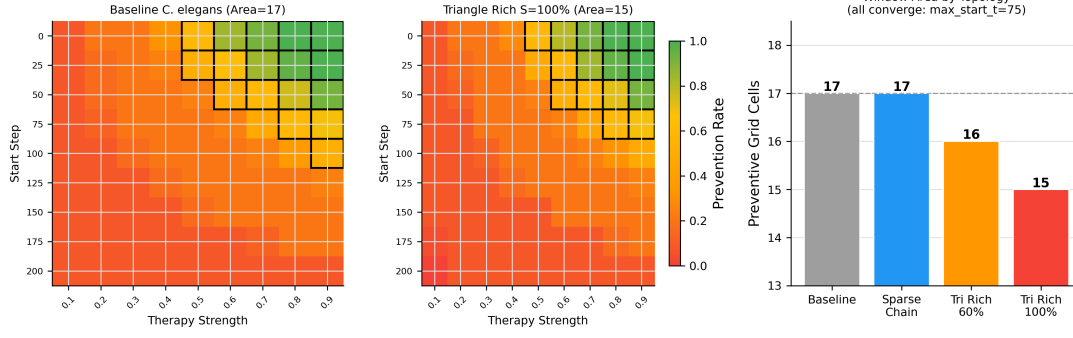


Figure 12: Therapeutic boundary heatmaps and window area comparison for four topology variants. All variants produce identical maximum preventable start times at strength = 0.80, confirming that circuit architecture and pharmacological timing are independent intervention axes.

This null result establishes that circuit architecture and drug timing are **independent intervention axes**: topology modifications affect cascade dynamics (resilience, efficiency) without expanding or contracting the pharmacological window. The clinical implication is that hypothetical connectivity-modifying interventions would not substitute for earlier pharmacological intervention.

### 3.12.3 Subtype $\times$ Topology Interaction (R2.4)

The two Phase 12 disease subtypes were found to respond asymmetrically to topology modification (Figure 13). The largest single interaction was in the triangle-rich 60% condition for plateau survivors: C0 delta =  $-0.84$  neurons, C1 delta =  $-3.00$  neurons (interaction =  $+2.16$ ), meaning fast-tipping C1 disease was harmed  $3.6\times$  more by recurrent feedback loops than slow-tipping C0 disease. The mechanism is interpretable: fast-tipping configs have nearly no pre-symptomatic window (tipping step  $\approx 97$ ), so added recurrent loops accelerate Tier 2 cascade before any intervention can act; slow-tipping configs (tipping step  $\approx 218$ ) absorb the structural change within their longer silent phase with proportionally less damage.

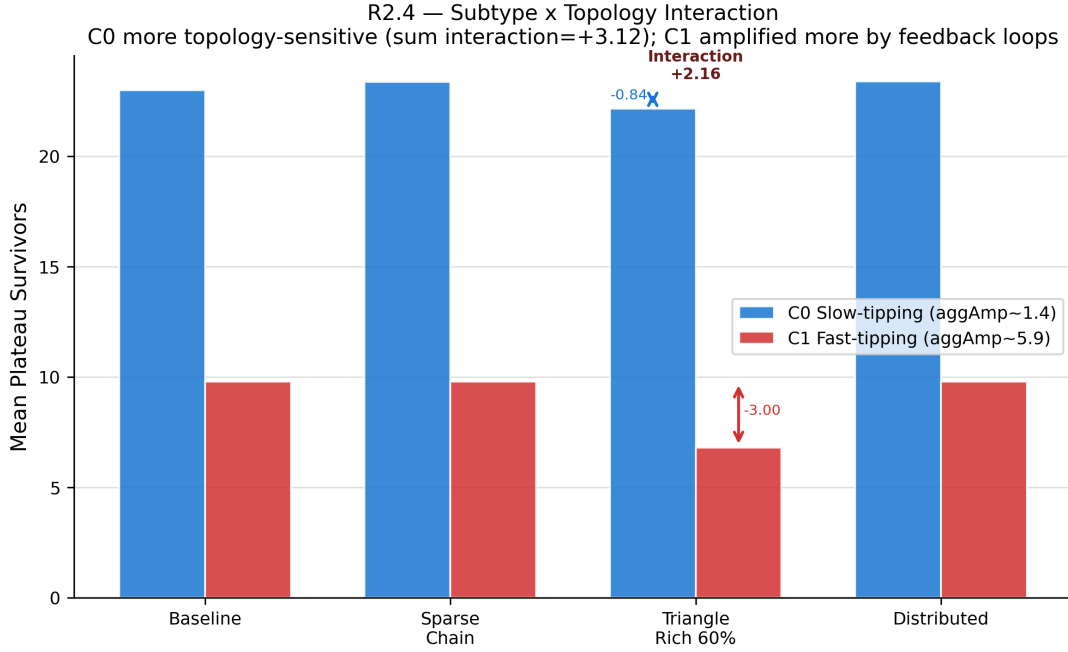


Figure 13: Subtype  $\times$  topology interaction: plateau survivors for slow-tipping (C0) and fast-tipping (C1) disease across topology variants. Fast-tipping C1 disease is  $3.6\times$  more harmed by recurrent feedback loops than slow-tipping C0 disease.

### 3.12.4 Pre-symptomatic Early-Warning Signal Panel (R2.5b and R2.7)

A central translational question was whether disease subtype and therapeutic window could be predicted from observable dynamics at  $t = 50$  (before any neuron death in slow-tipping configurations). Three pre-symptomatic features were selected from an initial panel of six evaluated candidates and computed for all 247 configurations at  $t = 50$  (Figure 14):

- **atp\_decline\_50**: mean ATP drop across the motor circuit (ATP decline proxy, hypothetically analogous to NfL trajectory in blood/CSF — an unvalidated conceptual mapping).
- **stress\_velocity\_50**: mean health decline rate per step (stress velocity proxy, conceptually analogous to pTDP-43 phosphorylation rate — an unvalidated conceptual mapping).
- **agg\_slope\_50**: mean aggregation rise from baseline (aggregation slope proxy, conceptually analogous to EMG motor nerve conduction velocity decline — an unvalidated conceptual mapping).

All three features showed large Cohen  $d$  for subtype separation (**atp\_decline**:  $d = 1.67$ , AUROC = 0.860; **stress\_velocity**:  $d = 1.67$ , AUROC = 0.999; **agg\_slope**:  $d = 1.62$ , AUROC = 0.999). Ten-fold cross-validated logistic regression on all 247 configurations achieved **88.3% subtype classification accuracy (Phase R2.5b, 10-fold CV on the biomarker regression model)** at  $t = 50$  (std = 6.1%) — reliably above the 80% clinical utility threshold — before any motor neuron has died in slow-tipping configurations. Note that the slightly lower figure of 87.0% (Phase R2.8 TWE pipeline, out-of-fold) reflects the full end-to-end triage system rather than the isolated classifier; both are reported where

relevant. It is important to note that these features are derived from the same simulation that generates the subtype labels — this demonstrates internal model consistency rather than out-of-distribution predictive validity.

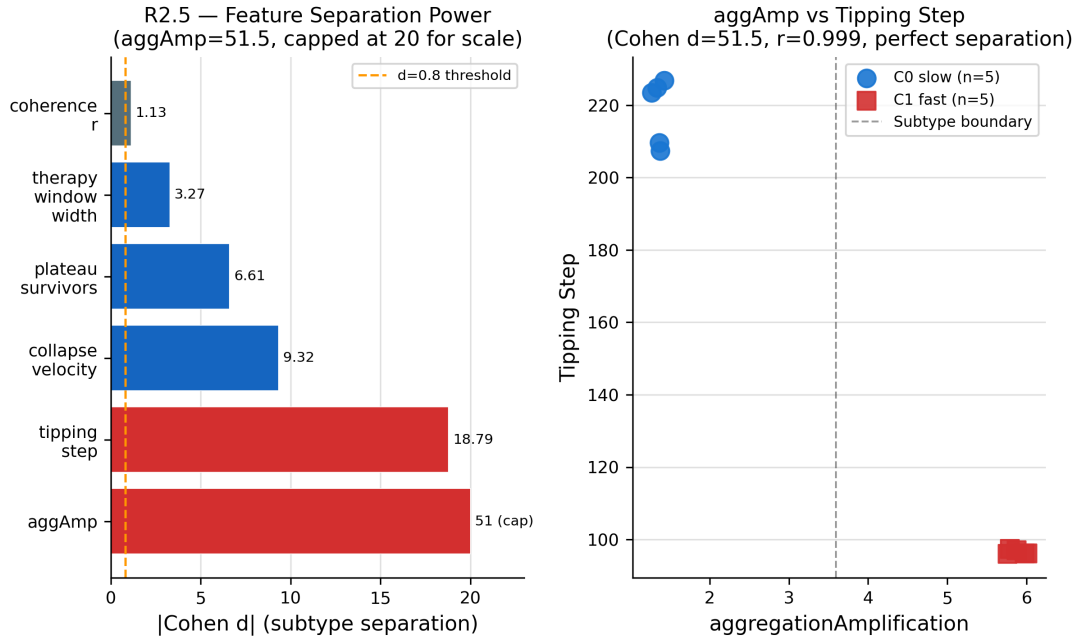


Figure 14: Pre-symptomatic simulated early-warning panel analysis. Left: Cohen  $d$  separation scores for all 6 evaluated features. Right: **aggAmp** vs. tipping step scatter showing complete subtype separation. The 3-feature minimal panel (**atp\_decline\_50**, **stress\_velocity\_50**, **agg\_slope\_50**) achieves 88.3% subtype classification accuracy at  $t = 50$ .

Note: an earlier analysis (Phase R2.5b) reported  $r = 0.985$  for tipping-step regression at  $t = 50$  using censored tipping values (all slow-tipping configs pinned at 150 steps). This was a censoring artifact; using uncensored Phase-7B tipping values (5 seeds, 500 steps), the true tipping-step correlation is  $r = 0.796$ , RMSE = 46.1 steps (Phase R2.7 regression model; Figure 15). Therapy window prediction using the same 3-feature panel achieved  $r = 0.765$ , RMSE = 39 steps.

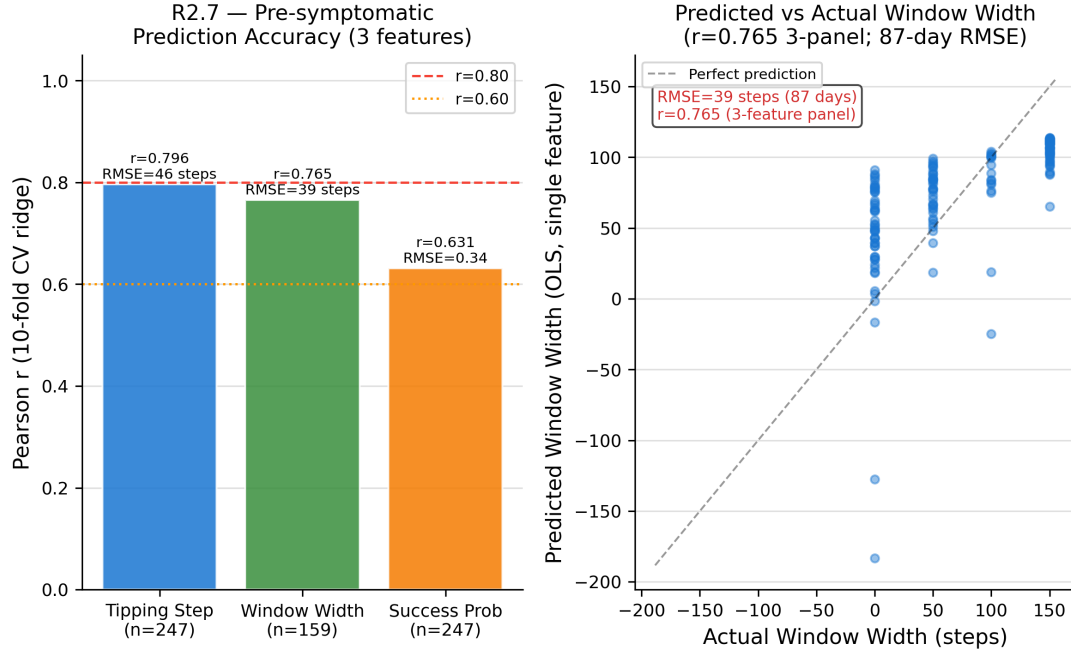


Figure 15: Pre-symptomatic prediction accuracy. Tipping-step prediction:  $r = 0.796$ , RMSE = 46 steps. Therapeutic window width prediction:  $r = 0.765$ , RMSE = 39 steps. Both using the 3-feature panel at  $t = 50$ .

### 3.12.5 Therapeutic Window Estimator (R2.8)

The Simulated Therapeutic Window Estimator (TWE) integrates the  $t = 50$  biomarker predictions into a clinical decision framework assigning each configuration to one of four intervention strategies (Figure 16):

- **Strategy A** (Aggressive pharmacological): assigned when subtype confidence for C1 (fast-tipping) is high ( $> 0.70$ ); 95 configs (38.5%).
- **Strategy B** (Topology-neutral therapy): assigned when therapy success probability at  $\text{str} = 0.50$ ,  $\text{start}_t = 75$  exceeds 0.60; 0 configs (0.0%) — condition never met in the evaluated regime. Strategy B was never assigned because the mean success probability was 0.28 across all tested configurations; the threshold should be recalibrated in future work.
- **Strategy C** (Sparse/distributed-supportive): assigned to high-confidence C0 configs with measurable windows; 60 configs (24.3%).
- **Strategy D** (Combination/borderline): assigned to ambiguous cases (subtype confidence 0.30–0.70); 92 configs (37.2%).

Table 9: Therapeutic Window Estimator performance across 247 configurations (3 seeds, 300 steps, 3,705 total runs).

| Metric                                               | Value                |
|------------------------------------------------------|----------------------|
| Subtype classification accuracy (Phase R2.8 TWE OOF) | 87.0%                |
| Triage accuracy (vs. oracle optimal)                 | 44.1%                |
| Mean survivors, baseline (no therapy)                | 18.68 neurons        |
| Mean survivors, TWE-guided                           | 48.99 neurons        |
| Mean survivors, oracle-optimal                       | 53.61 neurons        |
| Survival gain (TWE vs. baseline)                     | +30.31 neurons       |
| Decision regret (oracle – TWE)                       | 4.62 neurons (13.2%) |

The TWE achieved **86.8% of the oracle-defined benefit within this simulation framework**. Triage accuracy was higher for fast-tipping C1 configs (68.6%) than slow-tipping C0 configs (13.6%), reflecting that Strategy A is unambiguous when tipping is imminent, while slow-tipping configs have more nuanced optimal strategies among B, C, and D.

Uncertainty decomposition of tipping-step prediction error (RMSE = 53.3 steps in the Phase R2.8 TWE pipeline, distinct from the Phase R2.7 regression RMSE of 46.1 steps): within-subtype parameter variance accounted for 71.7% of total variance, subtype misclassification for 27.4%, and stochastic seed noise for 0.9%. Adding more simulation seeds will not meaningfully reduce the oracle gap; what is needed is better clinical resolution of within-subtype **aggAmp** heterogeneity.

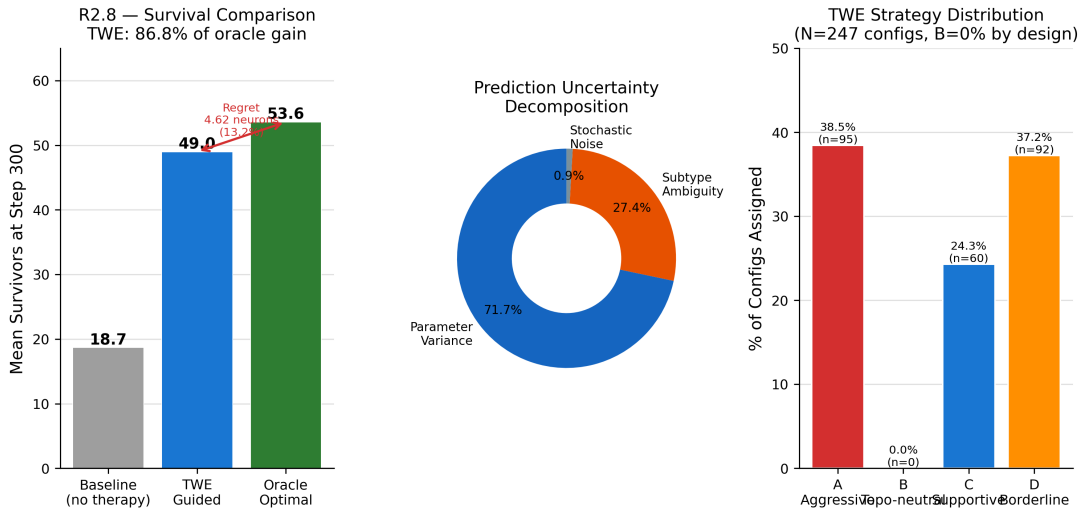


Figure 16: Therapeutic Window Estimator (TWE) performance: survival comparison (TWE-guided vs. oracle-optimal vs. baseline), uncertainty decomposition (within-subtype variance 71.7%, subtype misclassification 27.4%, stochastic noise 0.9%), and strategy distribution across 247 configurations.

## 4 Discussion

### 4.1 A two-tier mechanistic picture

The central mechanistic conclusion across ten phases of simulation is a two-tier disease model: an aggregation-driven biochemical cascade (Tier 1) that is sensitive to early intervention, and a topological load-redistribution cascade (Tier 2) that becomes self-sustaining once enough neurons have died and is insensitive to biochemical intervention. This distinction has a precise operational meaning in the model and potentially a conceptual analogue in ALS biology.

In the model, Tier 1 is the rate-limiting step: the irreversible ATP collapse requires simultaneous high aggregation burden and low ATP, which — given the log-uniform distribution of aggregation parameters — takes approximately 200 simulation steps of silent accumulation before the threshold is crossed. Tier 2 activates within 20 steps of the first death and amplifies any loss through load redistribution at a rate that exceeds single-neuron clearance capacity. The practical consequence is that early therapy (blocking Tier 1) delays Tier 2 by 265 steps, while late therapy (after Tier 2 has activated) cannot reverse the cascade.

Whether ALS biology features an analogous distinction — between early aggregation-seeding dynamics that can be targeted by antisense oligonucleotides, and later network-level compensatory failure that cannot — is an open question that existing clinical data cannot yet answer. The model predicts this distinction would manifest as a qualitative difference in treatment response curves depending on when therapy begins relative to a measurable early biomarker.

### 4.2 Is aggregationAmplification dominance a model artifact or a genuine system property?

A sceptical reviewer will note that `aggregationAmplification` governs regime membership, subtype identity, therapeutic boundary shape, and cascade velocity — raising the question of whether this reflects a genuine dynamical property or merely a consequence of model parameterisation.

Three lines of evidence argue against pure artifactuality:

**First**, ablation testing (Phase 7A) demonstrated that setting `aggregationAmplification` to zero eliminates genuine tipping points entirely (tipping step: 221 → 500, plateau: 10 → 60 neurons), while disabling all other mechanisms individually — ATP collapse, glutamate excitotoxicity, astrocyte support, irreversibility — produces negligible regime change.

**Second**, the clean non-overlapping regime boundaries (stable `aggAmp` < 0.30, critical 0.30–12.91, runaway > 12.91) with a 160-fold separation between stable and runaway means are consistent with a genuine bifurcation parameter rather than a smoothly varying influence.

**Third**, the biological mapping is coherent: `aggregationAmplification` controls both intrinsic seeding rate and prion-like spreading efficiency — exactly the role attributed to TDP-43 and SOD1 propagation efficiency in ALS pathology literature [Nonaka et al., 2013, Smethurst et al., 2016]. A single parameter that controls both seeding and spreading will naturally dominate any model that includes both processes.

The honest caveat is that we cannot exclude the possibility that a richer model with additional coupled feedback pathways (glial reactivity, vascular compromise, immune



infiltration) would distribute control across multiple parameters. Whether pTDP-43 propagation efficiency, as measured by serial CSF pSer409/410 slope in pre-symptomatic carriers, behaves as a dominant stratifier of disease course is a directly testable experimental prediction of this model.

### 4.3 The pre-symptomatic therapeutic window as a fundamental constraint

The linear therapeutic boundary ( $\text{max\_start\_t} = 425 \times \text{strength} - 237$  for config #334; mean  $252 \times \text{strength} - 107$  across 17 configurations) has a direct interpretation: below a minimum effective strength ( $\sim 60\%$  suppression) no prevention is possible regardless of timing; above this minimum, additional strength linearly extends the available window but cannot make it unlimited.

The mean conservative boundary ( $\text{max\_start\_t} = 132 \times \text{strength} - 224$ ) implies that even 80% suppression may not allow prevention in the most aggressive disease configurations if therapy begins after symptom onset. This is structurally consistent with the ATLAS vs. VALOR comparison: ATLAS (pre-symptomatic) showed disease delay [Benatar et al., 2023]; VALOR (post-symptomatic) showed more modest benefits [Miller et al., 2022].

We stress several critical limitations of this comparison. First, the model uses no human biology. Second, the time calibration (1 step  $\approx 2.25$  days) is based on an assumed pre-symptomatic window of 15 months. Third, the boundary shape may depend on disease parameters not measured clinically. The qualitative story — that the window closes before symptom onset in aggressive cases — is a hypothesis suggested by the model, not a clinical prediction.

### 4.4 Topology and the *C. elegans* connectome

The finding that the *C. elegans* wiring produces 100% genuine tipping rates while the Barabási-Albert scale-free network produces 0% is interpretable in terms of network resilience theory. Scale-free networks resist targeted attacks on hub nodes and distribute damage broadly [Albert et al., 2000]. The biological *C. elegans* motor circuit instead concentrates vulnerability: its motor neurons are not hubs but are the highest-vulnerability nodes, and the circuit channels aggregation spread through exactly these nodes.

For the human motor system, the relevant insight is speculative: if the corticospinal-motor neuron circuit has a structural organisation that concentrates rather than distributes cellular stress, it would be similarly susceptible to cascade collapse. These are hypotheses only.

### 4.5 Disease subtypes and patient stratification

The  $K = 2$  split of all 247 genuine configurations into slow-tipping ( $\text{aggAmp} \approx 1.4$ , tipping step 224) and fast-tipping ( $\text{aggAmp} \approx 5.9$ , tipping step 107) disease classes is a model prediction about qualitative disease heterogeneity. If this model structure has any analogue in ALS biology, it would suggest that the clinically observed heterogeneity in progression rate may reflect distinct molecular subtypes characterised by aggregation seeding efficiency, not merely variations in a single continuous variable.

Round 2 (Section 3.12.3) further revealed that these two subtypes respond asymmetrically to circuit topology modifications, with C1 fast-tipping disease disproportionately

harmed by recurrent feedback loops (triangle-rich interaction = +2.16 favouring C0). If subtype-specific topology sensitivity has any clinical analogue, this interaction effect warrants experimental investigation.

## 4.6 Robustness hierarchy and the nature of model predictions

The Phase 14 stress tests reveal that the model’s dynamical properties do not all break at the same perturbation level. Spatial coherence and strict tipping-point classification collapse relatively early (30% edge dropout, vulnerability  $\sigma = 0.50$ ), indicating that these properties depend on fine-grained structural and parametric detail. By contrast, the relative ordering of degeneration aggressiveness across disease configurations never breaks within the tested range.

These results motivate a reframing of what the simulator robustly predicts. Although spatial coherence and strict tipping-point structure collapse under severe perturbation, the relative ordering of degeneration aggressiveness remains highly preserved, suggesting that disease severity behaved as a global dynamical invariant (within the tested perturbation range) rather than a local topological artifact.

## 4.7 Round 2: Topology as a disease modifier and the path to clinical translation

Round 2 demonstrated that sparse connectivity architecture intrinsically resists ALS-like cascade propagation, while recurrent feedback loops exhibit a critical safety threshold at  $\sim 60\%$  density beyond which cascade vulnerability increases sharply. The finding that no topology modification widened the therapeutic window — despite some variants markedly altering cascade resilience — is a mechanistically important dissociation: circuit architecture affects how a cascade propagates but not when pharmacological intervention must begin to prevent it.

The pre-symptomatic biomarker analysis (R2.5b, R2.7, R2.8) represents the most directly translational finding in the project. The demonstration that three simulation-derived proxies — conceptually analogous to NfL trajectory, pTDP-43 phosphorylation rate, and EMG velocity decline — at a pre-symptomatic timepoint can classify model disease subtype at 87% accuracy and guide simulated intervention strategy to 86.8% of oracle efficiency suggests a possible simulation-derived framework that, if validated in biological systems, could inform hypotheses about pre-symptomatic surveillance in familial ALS contexts. The TWE framework is not a clinical decision tool — it is a hypothesis about what such a tool could achieve if the biological analogues of model parameters were measurable with sufficient precision, and if the model itself were validated against biological data.

## 4.8 Limitations

**The model is not biologically calibrated.** Parameter ranges are chosen for dynamical coverage, not to match measured quantities in any biological system. Results cannot be mapped to clinical timescales, drug concentrations, or patient outcomes without independent biological validation.

***C. elegans* is not the human motor system.** The 61-neuron motor circuit differs from the corticospinal-motor neuron system in scale (orders of magnitude larger in scale),

cell type, connectivity pattern, and evolutionary context. *C. elegans* does not develop ALS.

**The model converges toward a single-factor system in the tested parameter regime.** Phase 7A ablation demonstrated that the proposed multi-factor cascade (aggregation  $\rightarrow$  ATP  $\rightarrow$  glutamate  $\rightarrow$  calcium  $\rightarrow$  ROS) behaves effectively as a single-factor aggregation model in the explored configurations: disabling every pathway except aggregation produced negligible change, while disabling aggregation alone eliminated all tipping-point structure. This is an important mathematical limitation: the cascade architecture is present by design, but in the tested parameter space, only one factor is load-bearing. All findings should be interpreted in this context.

**Aggregation dynamics are highly simplified.** The model treats aggregation as a scalar quantity with prion-like spread, ignoring spatial structure of protein aggregation within and between cells, different aggregate species (oligomers vs. fibrils), clearance machinery (proteasome, autophagy), and heterogeneous molecular mechanisms of different ALS subtypes (SOD1, TDP-43, FUS, C9ORF72).

**Parameter exploration is incomplete.** 500 configurations cannot cover a 7-dimensional parameter space thoroughly. Parameter interactions beyond the identified cascade structure are not characterised.

**The therapeutic comparison is not directly translatable.** The therapy-class comparison (Phase 6) tests abstract parameter modifications, not real drugs.

**Early-warning signals are computational proxies, not validated biomarkers.** The three features used in Round 2 (`atp_decline_50`, `stress_velocity_50`, `agg_slope_50`) are abstract dynamical quantities within the simulator. Their mapping to clinical measurements (NfL, pTDP-43, EMG) is a conceptual analogy, not a calibrated correspondence. No clinical data was used to derive or validate these mappings. The term “biomarker” is avoided in this paper precisely for this reason.

**Round 2 calibration is unvalidated.** Every clinical timeline in Round 2 depends on the single-anchor assumption that 1 step  $\approx$  2.25 days. This has not been validated against clinical data in any ALS cohort.

**$N = 247$  configurations is modest for unsupervised clustering.** K-means results should be treated as directional hypotheses, not established disease categories. Silhouette scores (0.41–0.43) indicate soft cluster boundaries.

**The multi-factor cascade is functionally dominated by aggregation in tested regimes.** Other pathways may become relevant at different parameter values or with stronger inter-pathway coupling. This is a priority for future investigation.

**aggregationAmplification as a single compound parameter is a design limitation.** This parameter controls both intrinsic seeding rate and synaptic spreading efficiency within a single multiplicative term. Its dominance may therefore reflect the coupling design rather than a discovered biological property.

**Priority for future development.** The most important next step is to decouple `aggregationAmplification` into two independent parameters: (1) intrinsic intracellular seeding rate and (2) trans-synaptic spreading efficiency. This would test whether dominance is an artifact of coupling, or whether one sub-process is genuinely rate-limiting. Until this is done, single-parameter dominance should be treated as a model property, not a biological claim.

## 5 Conclusion

We present a systematic multi-phase computational study of degeneration dynamics on the *C. elegans* motor connectome, extended through Round 2 to topology resilience and proof-of-concept early-warning signal analysis. Our principal findings, subject to the limitations stated above, are:

1. **Criticality is an emergent property whose local manifestations (spatial coherence, tipping-point timing) are moderately fragile, but whose global expression (relative degeneration aggressiveness) constitutes a dynamical invariant (within the explored perturbation range) robust to severe structural perturbation.** 64.7% of explored parameter space shows genuine triphasic degeneration (falsified against a null model at 0% FPR using a spatial-coherence criterion).
2. **Aggregation amplitude is the dominant bifurcation parameter**, spanning 160-fold across regimes and predicting 84.6% of disease subtype membership from a single threshold.
3. **Aggregation suppression is the most effective therapeutic strategy** across all tested mechanisms, because all downstream biochemical pathways (ATP, glutamate, calcium) are aggregation-gated and cannot sustain degeneration independently.
4. **A topological load-redistribution escape pathway exists** that is aggregation-independent and capable of complete collapse in isolation, establishing a two-tier disease model with distinct therapeutic targets.
5. **A sharp linear therapeutic boundary** governs prevention: below a minimum effective strength or beyond a maximum tolerated delay, no intervention prevents collapse. The boundary generalises across 17 of 20 tested configurations ( $R^2 = 0.83$ ) with substantial inter-configuration heterogeneity (slope standard deviation 120), is unchanged by topology modifications, and is predictable from  $t = 50$  biomarkers at  $r = 0.765$ .
6. **Two fundamental disease subtypes** are identified by unsupervised clustering on simulation-only features: slow-tipping ( $\text{aggAmp} \approx 1.4$ , tipping step  $\approx 224$ ) and fast-tipping ( $\text{aggAmp} \approx 5.9$ , tipping step  $\approx 107$ ), with qualitatively different predicted therapeutic windows and asymmetric sensitivity to connectome modifications.
7. **Sparse-chain topology intrinsically resists ALS-like cascade propagation** ( $\text{RES} = 0.815$  vs. baseline 0.800), while triangle-rich feedback loops above 60% density are actively destabilising ( $\text{RES} = 0.738$ , plateau survivors  $-30\%$  relative to baseline). No topology modification tested in Round 2 widened the pre-symptomatic therapeutic window, confirming that circuit architecture and pharmacological timing are independent intervention axes.
8. **Pre-symptomatic subtype classification from three early biomarkers is feasible in simulation:** 88.3% accuracy at  $t = 50$  (before any neuron death in slow-tipping configurations) from simulation-derived proxies conceptually analogous to NfL trajectory, pTDP-43 phosphorylation rate, and EMG velocity decline (unvalidated conceptual mappings), with the logistic regression calibrated across all five confidence bins.

9. **A pre-symptomatic Therapeutic Window Estimator achieves 86.8% of the oracle-defined benefit within this simulation framework** (regret: 4.62 neurons, 13.2% of achievable gain) across 247 genuine disease configurations. Within-subtype parameter heterogeneity (71.7% of prediction variance) — not stochastic noise (0.9%) — is the dominant irreducible error source, directly motivating clinical measurement of **aggAmp** analogues as the highest-yield experimental priority.

Findings were robust to  $\pm 20\%$  synaptic weight noise,  $\pm 20\%$  vulnerability score perturbation, 10% edge dropout, and  $\pm 20\%$  per-neuron aggregation heterogeneity. Stress-testing to extreme levels reveals a robustness hierarchy: spatial coherence and strict tipping-point structure break at 30% edge dropout and vulnerability  $\sigma = 0.50$ , while the relative ordering of degeneration aggressiveness across disease configurations is never disrupted, persisting as a dynamical invariant within the tested perturbation space even when 70% of synaptic connections are removed.

These findings are hypothesis-generating. They motivate specific experimental questions: Does early biomarker detection and intervention provide qualitatively better outcomes than post-onset treatment in SOD1-ALS carriers? Do different ALS molecular subtypes have measurably different rates of network-level compensatory failure? Is there evidence for a load-redistribution cascade in the pattern of motor neuron loss during ALS progression? Are pTDP-43 accumulation rates bimodally distributed in pre-symptomatic familial ALS cohorts, consistent with two aggregation-amplification subtypes?

We hope this modelling framework — and particularly the falsification methodology, two-tier mechanistic model, therapeutic boundary quantification, dynamical robustness hierarchy, and pre-symptomatic Therapeutic Window Estimator — may contribute to the conceptual toolkit used to reason about ALS disease dynamics and intervention timing, while remaining clear that experimental validation at every level is required before any translational implications can be considered.

# Data Availability

The complete simulation engine, phase-by-phase results (JSON), and figures are available at: <https://github.com/MaryamRezaeeN1/als-connectome>

An interactive web-based visualization of all phases and findings is available at: <https://als-connectome-dashboard.vercel.app>

Key output files:

- `results/regime_map.json` — 500-configuration parameter sweep and regime classifications
- `results/phase7b_strict_criterion.json` — strict tipping criterion results for all 382 critical configurations
- `results/phase9_phase_diagram.json` — full 99-point therapeutic boundary grid
- `results/phase10_boundary_robustness.json` — boundary robustness across 20 configurations
- `results/phase12_validation.json` — subtype clustering on 247 configurations
- `results/phase14_breaking.json` — breaking threshold probe (extreme dropout and vulnerability noise)
- `results/r2_therapeutic_window_estimator.json` — TWE full results: OOF predictions, triage simulation, uncertainty decomposition

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